

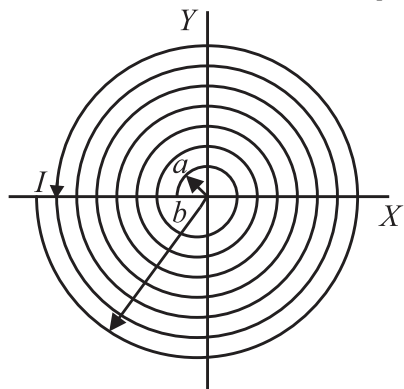
Lakshya JEE 2027

KPP-13

Moving Charges and Magnetism

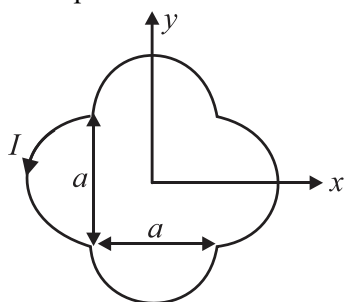
By: Saleem Bhaiya

1. A long insulated copper wire is closely wound as a spiral of N turns. The spiral has inner radius a and outer radius b . The spiral lies in the X - Y plane and a steady current I flows through the wire. The Z -component of the magnetic field at the centre of the spiral is [IIT-JEE-2011]



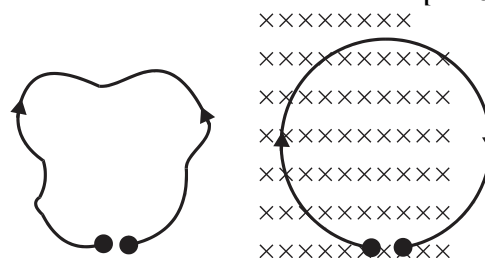
- (1) $\frac{\mu_0 NI}{2(b-a)} \ln\left(\frac{b}{a}\right)$ (2) $\frac{\mu_0 NI}{2(b-a)} \ln\left(\frac{b+a}{b-a}\right)$
 (3) $\frac{\mu_0 NI}{2b} \ln\left(\frac{b}{a}\right)$ (4) $\frac{\mu_0 NI}{2b} \ln\left(\frac{b+a}{b-a}\right)$

2. A loop carrying current ' I ' lies in the x - y plane as shown in the figure. The unit vector $\hat{k}$ is coming out of the plane of the paper. The magnetic moment of the current loop is [JEE-2012]



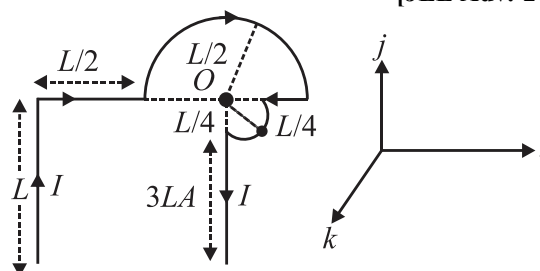
- (1) $a^2 I \hat{k}$
 (2) $\left(\frac{\pi}{2} + 1\right) a^2 I \hat{k}$
 (3) $-\left(\frac{\pi}{2} + 1\right) a^2 I \hat{k}$
 (4) $(2\pi + 1) a^2 I \hat{k}$

3. A thin flexible wire of length L is connected to two adjacent fixed points and carries a current I in the clockwise direction, as shown in the figure. When the system is put in a uniform magnetic field of strength B going into the plane of the paper, the wire takes the shape of a circle. The tension in the wire is [IIT-JEE-2010]



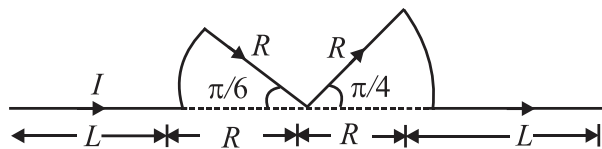
- (1) IBL
 (2) $\frac{IBL}{\pi}$
 (3) $\frac{IBL}{2\pi}$
 (4) $\frac{IBL}{4\pi}$

4. Which one of the following options represents the magnetic field $\vec{B}$ at O due to the current flowing in the given wire segments lying on the xy plane? [JEE Adv.-2022]



- (1) $\vec{B} = \frac{-\mu_0 I}{L} \left(\frac{3}{2} + \frac{1}{4\sqrt{2}\pi} \right) \hat{k}$
 (2) $\vec{B} = -\frac{\mu_0 I}{L} \left(\frac{3}{2} + \frac{1}{2\sqrt{2}\pi} \right) \hat{k}$
 (3) $\vec{B} = \frac{-\mu_0 I}{L} \left(1 + \frac{1}{4\sqrt{2}\pi} \right) \hat{k}$
 (4) $\vec{B} = \frac{-\mu_0 I}{L} \left(1 + \frac{1}{4\pi} \right) \hat{k}$

5. A conductor (shown in the figure) carrying constant current I is kept in the x - y plane in a uniform magnetic field $\vec{B}$. If F is the magnitude of the total magnetic force acting on the conductor, then the correct statement(s) is (are) [JEE Adv.-2015]



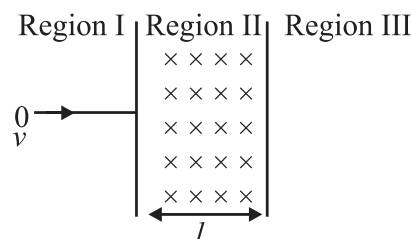
- (1) If $\vec{B}$ is along $\hat{z}$, $F \propto (L + R)$
 - (2) If $\vec{B}$ is along $\hat{x}$, $F = 0$
 - (3) If $\vec{B}$ is along $\hat{y}$, $F \propto (L + R)$
 - (4) If $\vec{B}$ is along $\hat{z}$, $F = 0$
6. Two parallel wires in the plane of the paper are distance x_0 apart. A point charge is moving with speed u between the wires in the same plane at a distance x_1 from one of the wires. When the wires carry current of magnitude I in the same direction, the radius of curvature of the path of the point charge is R_1 . In contrast, if the currents I in the two wires have directions opposite to each other, the radius of curvature of the path is R_2 . If $\frac{x_0}{x_1} = 3$, and value of $\frac{R_1}{R_2}$ is [JEE Adv.-2014]

7. An electron and a proton are moving on straight parallel paths with same velocity. They enter a semi-infinite region of uniform magnetic field perpendicular to the velocity. Which of the following statement(s) is/are true? [IIT-JEE-2011]

- (1) They will never come out of the magnetic field region
- (2) They will come out travelling along parallel paths
- (3) They will come out at the same time
- (4) They will come out at different times

8. An α -particle (mass 4 amu) and a singly charged sulfur ion (mass 32 amu) are initially at rest. They are accelerated through a potential V and then allowed to pass into a region of uniform magnetic field which is normal to the velocities of the particles. Within this region, the α -particle and the sulfur ion move in circular orbits of radii r_α and r_s , respectively. The ratio (r_s/r_α) is _____. [JEE Adv.-2021]

9. A particle of mass m and charge q , moving with velocity v enters Region II normal to the boundary as shown in the figure. Region II has a uniform magnetic field B perpendicular to the plane of the paper. The length of the Region II is l . Choose the correct choice (s) [IIT-JEE-2008]

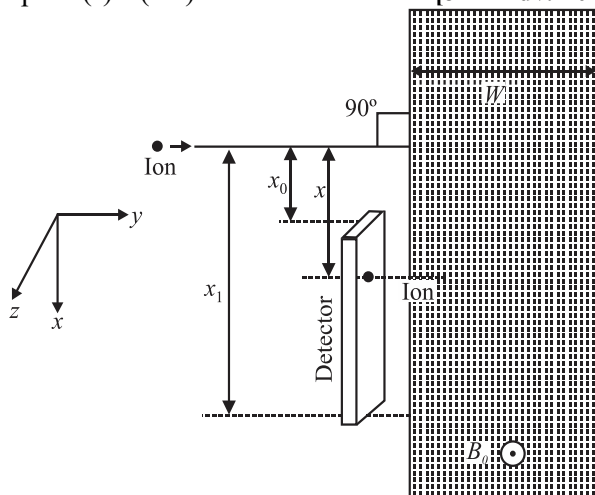


- (1) The particle enters Region III only if its velocity $v > \frac{qBl}{m}$
 - (2) The particle enters Region III only if its velocity $v < \frac{qBl}{m}$
 - (3) Path length of the particle in Region II is maximum when velocity $v = qBl/m$
 - (4) Path spent in Region II is same for any velocity v as long as the particle returns to Region I
10. Consider the motion of a positive point charge in a region where there are simultaneous uniform electric and magnetic fields $E = E_0\hat{j}$ and $B = B_0\hat{j}$. At time $t = 0$, this charge has velocity v in the x - y plane, making an angle θ with the x -axis. Which of the following option(s) is/are correct for time $t > 0$? [IIT-JEE-2012]
- (1) If $\theta = 0^\circ$, the charge moves in a circular path in the x - z plane
 - (2) If $\theta = 0^\circ$, the charge undergoes helical motion with constant pitch along the y -axis
 - (3) If $\theta = 10^\circ$, the charge undergoes helical motion with its pitch increasing with time, along the y -axis
 - (4) If $\theta = 90^\circ$, the charge undergoes linear but accelerated motion along the y -axis

11. A particle of mass M and positive charge Q , moving with a constant velocity $\vec{u}_1 = 4\hat{i} \text{ ms}^{-1}$, enters a region of uniform static magnetic field normal to the x - y plane. The region of the magnetic field extends from $x = 0$ to $x = L$ from all values of y . After passing through this region, the particle emerges on the other side after 10 milliseconds with a velocity $\vec{u}_2 = 2(\sqrt{3}\hat{i} + \hat{j}) \text{ ms}^{-1}$. The correct statement(s) is (are): [IIT-JEE-2013]

- (1) The direction of the magnetic field is $-z$ direction
- (2) The direction of the magnetic field is $+z$ direction
- (3) The magnitude of the magnetic field $\frac{50\pi M}{3Q}$ units
- (4) The magnitude of the magnetic field is $\frac{100\pi M}{3Q}$ units

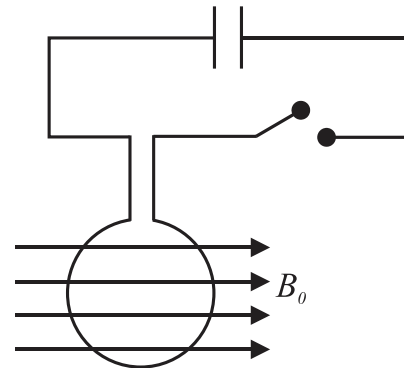
12. A positive, singly ionized atom of mass number A_M is accelerated from rest by the voltage 192 V. Thereafter, it enters a rectangular region of width w with magnetic field $\vec{B}_0 = 0.1\hat{k}$ Tesla, as shown in the figure. The ion finally hits a detector at the distance x below its starting trajectory. [Given: Mass of neutron/proton = $(5/3) \times 10^{-27}$ kg, charge of the electron = 1.6×10^{-19} C]. Which of the following option(s) is(are) correct? [JEE Adv.-2014]



- (1) The value of x for H^+ ion is 4 cm
- (2) The value of x for an ion with $A_M = 144$ is 48 cm
- (3) For detecting ions with $1 \leq A_M \leq 196$, the minimum height $(x_1 - x_0)$ of the detector is 55 cm
- (4) The minimum width w of the region of the magnetic field for detecting ions with $A_M = 196$ is 56 cm

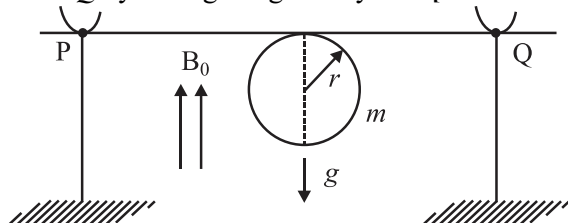
13. A steady current I goes through a wire loop PQR having shape of a right angle triangle with $PQ = 3x$, $PR = 4x$ and $QR = 5x$. If the magnitude of the magnetic field P due to this loop is $k\left(\frac{\mu_0 I}{48\pi x}\right)$, Find the value of k [IIT-JEE-2009]

14. A circular coil of radius R and N turns has negligible resistance. As shown in the schematic figure, its two ends are connected to two wires and it is hanging by those wires with its plane being vertical. The wires are connected to a capacitor with charge Q through a switch. The coil is in a horizontal uniform magnetic field B parallel to the plane of the coil. When the switch is closed, the capacitor gets discharged through the coil in a very short time. By the time the capacitor is discharged fully, magnitude of the angular momentum gained by the coil will be (assume that the discharge time is so short that the coil has hardly rotated during this time) [JEE-Adv.-2020]

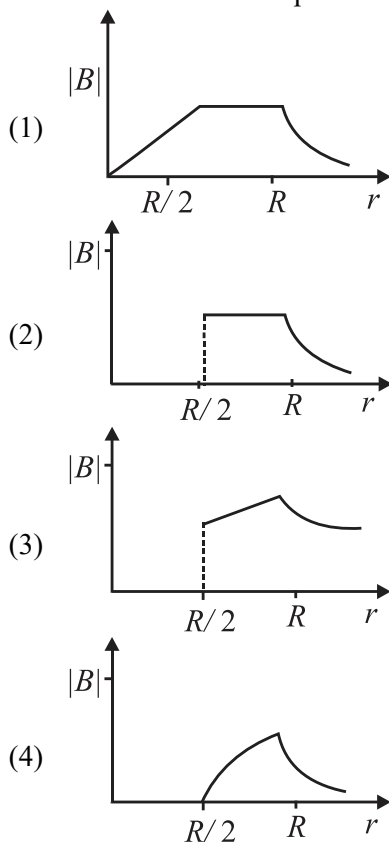


- (1) $\frac{\pi}{2} NQB_0 R^2$
- (2) $\pi NQB_0 R^2$
- (3) $2\pi NQB_0 R^2$
- (4) $4\pi NQB_0 R^2$

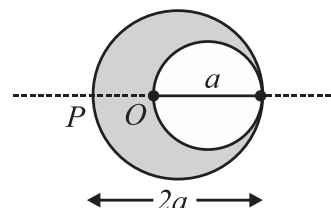
15. A thin stiff insulated metal wire is bent into a circular loop with its two ends extending tangentially from the same point of the loop. The wire loop has mass m and radius r and it is in a uniform vertical magnetic field B_0 , as shown in the figure. Initially, it hangs vertically downwards, because of acceleration due to gravity g , on two conducting supports at P and Q . When a current I is passed through the loop, the loop turns about the line PQ by an angle θ given by [JEE-Adv.-2024]



- (1) $\tan\theta = \pi r I B_0 / (mg)$
 (2) $\tan\theta = 2\pi r I B_0 / (mg)$
 (3) $\tan\theta = \pi r I B_0 / (2mg)$
 (4) $\tan\theta = mg / (\pi r I B_0)$
16. An infinitely long hollow conducting cylinder with inner radius $R/2$ and outer radius R carries a uniform current density along its length. The magnitude of the magnetic field, $|B|$ as a function of the radial distance r from the axis is best represented by [IIT-JEE-2012]



17. A cylindrical cavity of diameter a exists inside a cylinder of diameter $2a$ as shown in the figure. Both the cylinder and the cavity are infinitely long. A uniform current density J flows along the length. If the magnitude of the magnetic field at the point P is given by $\frac{N}{12} \mu_0 a J$, then the value of N is:



- Direction (Q. 18-20):** A charged particle (electron or proton) is introduced at the origin ($x = 0, y = 0, z = 0$) with a given initial velocity $\vec{v}$. A uniform electric field $\vec{E}$ and a uniform magnetic field $\vec{B}$ exist everywhere. The velocity $\vec{v}$, electric field $\vec{E}$ and magnetic field $\vec{B}$ are given in columns I, II and III, respectively. The quantities E_0, B_0 are positive in magnitude [JEE Adv.-2017]

	Column-I		Column-II		Column-III
I	Electron with $\vec{v} = 2 \frac{E_0}{B_0} \hat{x}$	i	$\vec{E} = E_0 \hat{z}$	P	$\vec{B} = -B_0 \hat{x}$
II	Electron with $\vec{v} = \frac{E_0}{B_0} \hat{y}$	ii	$\vec{E} = -E_0 \hat{y}$	Q	$\vec{B} = B_0 \hat{x}$
III	Proton with $\vec{v} = 0$	iii	$\vec{E} = -E_0 \hat{x}$	R	$\vec{B} = B_0 \hat{y}$
IV	Proton with $\vec{v} = 2 \frac{E_0}{B_0} \hat{x}$	iv	$\vec{E} = E_0 \hat{x}$	S	$\vec{B} = B_0 \hat{z}$

18. In which case will the particle move in a straight line with constant velocity?
 (1) (II) (iii) (S) (2) (IV) (i) (S)
 (3) (III) (ii) (R) (4) (III) (iii) (P)
19. In which case will the particle describe a helical path with axis along the positive z -direction?
 (1) (II) (ii) (R) (2) (IV) (ii) (R)
 (3) (IV) (i) (S) (4) (III) (iii) (P)
20. In which case would the particle move in a straight line along the negative direction of y -axis (i.e., move along $-\hat{y}$)?
 (1) (IV) (ii) (S) (2) (III) (ii) (P)
 (3) (II) (iii) (Q) (4) (III) (ii) (R)

21. In a particular system of units, a physical quantity can be expressed in terms of the electric charge e , electron mass m_e , Planck's constant h , and Coulomb's constant $k = \frac{1}{4\pi\epsilon_0}$, where ϵ_0 is the permittivity of vacuum. In terms of these physical constants, the dimension of the magnetic field is $[B] = [e]^\alpha [m_e]^\beta [h]^\gamma [k]^\delta$. The value of $\alpha + \beta + \gamma + \delta$ is _____. [JEE Adv.-2022]

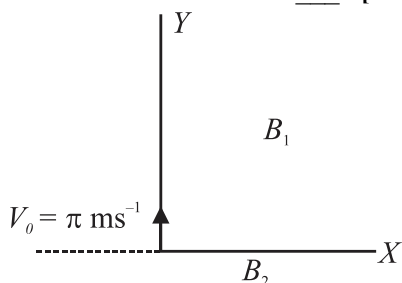
22. A dimensionless quantity is constructed in terms of electronic charge e , permittivity of free space ϵ_0 , Planck's constant h , and speed of light c . If the dimensionless quantity is written as $e^\alpha \epsilon_0^\beta h^\gamma c^\delta$ and n is a non-zero integer, then $(\alpha, \beta, \gamma, \delta)$ is given by [JEE Adv.-2024]

- (1) $(2n, -n, -n, -n)$ (2) $(n, -n, -2n, -n)$
(3) $(n, -n, -n, -2n)$ (4) $(2n, -n, -2n, -2n)$

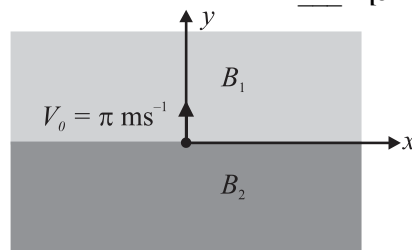
23. An infinitely long wire, located on the z -axis, carries a current I along the $+z$ -direction and produces the magnetic field $\vec{B}$. The magnitude of the line integral $\int \vec{B} \cdot d\vec{l}$ along a straight line from the point $(-\sqrt{3}a, a, 0)$ to $(a, a, 0)$ is given by [JEE Adv.-2024]

- (1) $7\mu_0 I/24$ (2) $7\mu_0 I/12$
(3) $\mu_0 I/8$ (4) $\mu_0 I/6$

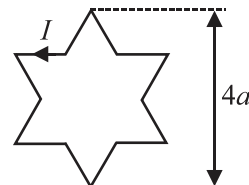
24. In the xy -plane, the region $y > 0$ has a uniform magnetic field $B_1 \hat{k}$ and the region $y < 0$ has another uniform magnetic field $B_2 \hat{k}$. A positively charged particle is projected from the origin along the positive Y -axis with speed $v_0 = \pi \text{ ms}^{-1}$ at $t = 0$, as shown in figure. Neglect gravity in this problem. Let $t = T$ be the time when the particle crosses the X -axis from below for the first time. If $B_2 = 4B_1$, the average speed of the particle, in ms^{-1} , along the X -axis in the time interval T is _____. [JEE Adv.-2018]



25. In the xy -plane, the region $y > 0$ has a uniform magnetic field $B_1 \hat{k}$ and the region $y < 0$ has another uniform magnetic field $B_2 \hat{k}$. A positively charged particle is projected from the origin along the positive y -axis with speed $v_0 = \pi \text{ ms}^{-1}$ at $t = 0$, as shown in figure. Neglect gravity in this problem. Let $t = T$ be the time when the particle crosses the x -axis from below for the first time. If $B_2 = 4B_1$, the average speed of the particle, in ms^{-1} , along the x -axis in the time interval T is _____. [JEE Adv.-2018]



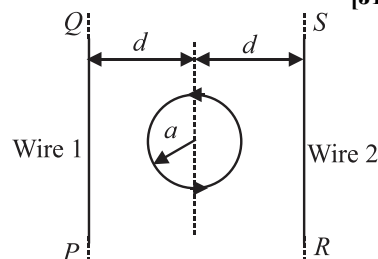
26. A symmetric star shaped conducting wire loop is carrying a steady state current I as shown in the figure. The distance between the diametrically opposite vertices of the star is $4a$. The magnitude of the magnetic field at the center of the loop is: [JEE Adv.-2017]



- (1) $\frac{\mu_0 I}{4\pi a} 3[\sqrt{3}-1]$ (2) $\frac{\mu_0 I}{4\pi a} 6[\sqrt{3}-1]$
(3) $\frac{\mu_0 I}{4\pi a} 6[\sqrt{3}+1]$ (4) $\frac{\mu_0 I}{4\pi a} 3[2-\sqrt{3}]$

Paragraph for Question 27 and 28:

The figure shows a circular loop of radius a with two long parallel wires (numbered 1 and 2) all in the plane of the paper. The distance of each wire from the center of the loop is d . The loop and the wires are carrying the same current I . The current in the loop is in the counterclockwise direction if seen from above. [JEE Adv.-2014]



27. When $d \approx a$ but wires are not touching the loop, it is found that the net magnetic field on the axis of the loop is zero at a height h above the loop. In that case
- (1) current in wire 1 and wire 2 is the direction PQ and RS , respectively $h \approx a$
 - (2) current in wire 1 and wire 2 is the direction PQ and SR , respectively $h \approx a$
 - (3) current in wire 1 and wire 2 is the direction PQ and SR , respectively $h \approx 1.2 a$
 - (4) current in wire 1 and wire 2 is the direction PQ and RS , respectively $h \approx 1.2 a$

28. Consider $d \gg a$, and the loop is rotated about its diameter parallel to the wires by 30° from the position shown in the figure. If the currents in the wires are in the opposite directions, the torque on the loop at its new position will be (assume that the net field due to the wires is constant over the loop)

- (1) $\frac{\mu_0 I^2 a^2}{d}$
- (2) $\frac{\mu_0 I^2 a^2}{2d}$
- (3) $\frac{\sqrt{3}\mu_0 I^2 a^2}{d}$
- (4) $\frac{\sqrt{3}\mu_0 I^2 a^2}{2d}$

29. A steady current I flows along an infinitely long hollow cylindrical conductor of radius R . This cylinder is placed coaxially inside an infinite solenoid of radius $2R$. The solenoid has n turns per unit length and carries a steady current I . Consider a point P at a distance r from the common axis. The correct statements (is/are)

[JEE Adv.-2013]

- (1) In the region $0 < r < R$, the magnetic field is non-zero
- (2) In the region $R < r < 2R$, the magnetic field is along the common axis
- (3) In the region $R < r < 2R$, the magnetic field is tangential to the circle of radius r , centered on the axis
- (4) In the region $r > 2R$, the magnetic field is non-zero

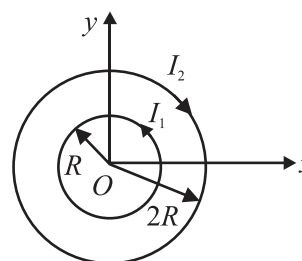
30. Two infinitely long straight wires lie in the xy -plane along the lines $x = \pm R$. The wire located at $x = +R$ carries a constant current I_1 and the wire located at $x = -R$ carries a constant current I_2 . A circular loop of radius R is suspended with its center at $(0, 0, \sqrt{3}R)$ and in a plane parallel to the xy -plane. This loop carries a constant current I in the clockwise direction as seen from above the loop. The current in the wire is taken to be positive, if it is in the $+\hat{j}$ -direction. Which of the following statements regarding the magnetic field B is/are true?

[JEE Adv.-2018]

- (1) If $I_1 = I_2$, then B cannot be equal to zero at the origin $(0, 0, 0)$
- (2) If $I_1 > 0$ and $I_2 < 0$ then B can be equal to zero at the origin $(0, 0, 0)$
- (3) If $I_1 < 0$ and $I_2 > 0$ then B can be equal to zero at the origin $(0, 0, 0)$
- (4) If $I_1 = I_2$, then the z -component of the magnetic field at the center of the loop is $\left(-\frac{\mu_0 I}{2R}\right)$

31. Two concentric circular loops, one of radius R and the other of radius $2R$, lie in the xy -plane with the origin as their common center, as shown in the figure. The smaller loop carries current I_1 in the anti-clockwise direction and the larger loop carries current I_2 in the clockwise direction, with $I_2 > 2I_1$. $\vec{B}(x, y)$ denotes the magnetic field at a point (x, y) in the xy -plane. Which of the following statement(s) is/are correct?

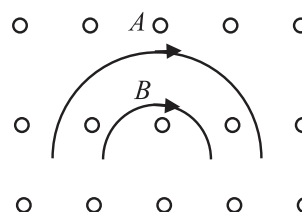
[JEE Adv.-2021]



- (1) $\vec{B}(x, y)$ is perpendicular to the xy -plane at any point in the plane
- (2) $|\vec{B}(x, y)|$ depends on x and y only through the radial distance $r = \sqrt{x^2 + y^2}$
- (3) $|\vec{B}(x, y)|$ is non-zero at all points for $r < R$
- (4) $\vec{B}(x, y)$ points normally outward from the plane for all the points between the two loops

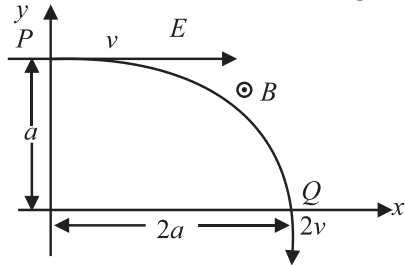
32. Two particles A and B of masses m_A and m_B respectively and having the same charge are moving in a plane. A uniform magnetic field exists perpendicular to this plane. The speeds of the particles are v_A and v_B respectively and the trajectories are as shown in the figure. Then

[IIT-JEE-2001]



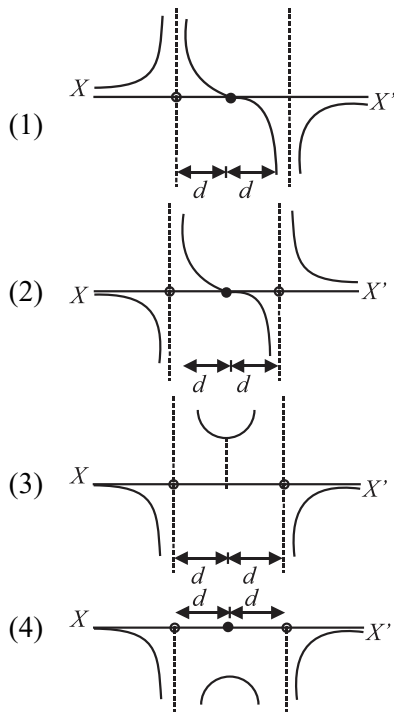
- (1) $m_A v_A < m_B v_B$
- (2) $m_A v_A > m_B v_B$
- (3) $m_A < m_B$ and $v_A < v_B$
- (4) $m_A = m_B$ and $v_A = v_B$

33. A particle of charge $+q$ and mass m moving under the influence of a uniform electric field $E\hat{i}$ and uniform magnetic field $B\hat{k}$ follows a trajectory from P to Q as shown in figure. The velocities at P and Q are $v\hat{i}$ and $-2v\hat{j}$. Which of the following statement(s) is/are correct? [JEE Main-2022]



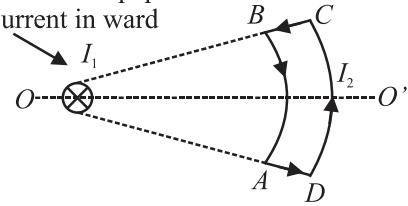
- (1) $E = \frac{3}{4} \left[\frac{mv^2}{qa} \right]$
- (2) Rate of work done by the electric field at P is $\frac{3}{4} \left[\frac{mv^3}{a} \right]$
- (3) Rate of work done by the electric field at P is zero
- (4) Rate of work done by the electric field at Q is zero

34. Two long parallel wires are at a distance $2d$ apart. They carry steady equal current flowing out of the plane of the paper as shown. The variation of the magnetic field B along the line XX' is given by [IIT-JEE-2000]

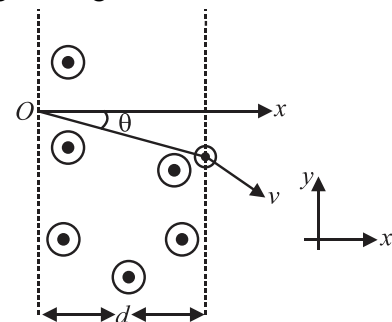


35. Which of the following statement is/are correct in the given figure? [IIT-JEE-2006]

Infinitely long wire kept perpendicular to the paper carrying current in ward



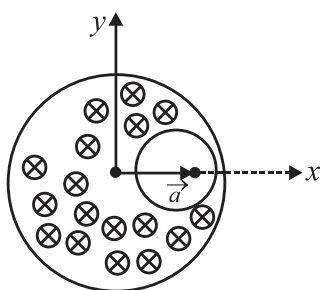
- (1) Net force on the loop is zero
 - (2) Net torque on the loop is zero
 - (3) Loop will rotate clockwise about axis OO' when seen from O
 - (4) Loop will rotate anticlockwise about OO' when seen from O
36. A proton moving with a constant velocity passes through a region of space without any change in its velocity. If E and B represent the electric and magnetic fields respectively, this region of space cannot be [IIT-JEE-1985]
- (1) $E = 0, B = 0$
 - (2) $E = 0, B \neq 0$
 - (3) $E \neq 0, B = 0$
 - (4) $E \neq 0, B \neq 0$
37. A particle of charge q is projected with a momentum $\vec{P} = P\hat{i}$ in the given region of magnetic field $\vec{B} = B\hat{k}$. It emerges from the magnetic field after deviating through an angle $\theta = 30^\circ$.



- (1) The value of $\vec{P}$ is $2qBd$
- (2) The value of $\vec{P}$ is qBd
- (3) Maximum change in momentum takes place for $d \geq \frac{P}{qB}$
- (4) Maximum change in momentum takes place for $d \geq \frac{P}{2qB}$

38. A charge particle is moving inside both electric field and magnetic field without any acceleration. Electric field is $10^3 \hat{i} \frac{N}{C}$ and magnetic field is $2 \times 10^{-2} \hat{j} T$. The velocity of charge particle is
- (1) $5 \times 10^4 \hat{k}$
 - (2) $2 \times 10^3 \hat{k}$
 - (3) $4 \times 10^4 (-\hat{k})$
 - (4) May be both $4 \times 10^4 \hat{k}$ and $4 \times 10^4 (-\hat{k})$ depend on nature of charge

39. Figure shows cross-section view of a infinite cylindrical wire with a cylindrical cavity, current density is uniform $\vec{J} = -j_0 \hat{k}$ as shown in figure.



- (1) Field inside cavity is uniform
 - (2) Field inside cavity is along $\vec{a}$
 - (3) Field inside cavity is perpendicular to $\vec{a}$
 - (4) If an electron is projected with velocity $v_0 \hat{j}$ it will move undeviated before colliding with cavity wall
40. A tightly-wound, long solenoid carries a current of 2.00 A. An electron is found to execute a uniform circular motion inside the solenoid with a frequency of 1.00×10^8 rev/s. Find the number of turns per metre in the solenoid.
- (1) 500 Turns/m
 - (2) 1020 Turns/m
 - (3) 1232 Turns/m
 - (4) 1420 Turns/m

41. A charged particle with some initial velocity is projected in a region where uniform electric and/or magnetic fields are present. In Column-I information about the existence of electric and/or magnetic field and direction of initial velocity of charged particle are given, while in column-II the possible paths of charged particle is mentioned. Match the entries of Column-I with the entries of Column-II.

Column-I		Column-II	
A	$\vec{E} = 0, \vec{B} \neq 0$ and initial velocity is at an unknown angle with $\vec{B}$	P	Straight line
B	$\vec{E} \neq 0, \vec{B} = 0$ and initial velocity is at an unknown angle with $\vec{E}$	Q	Parabola
C	$\vec{E} \neq 0, \vec{B} \neq 0, \vec{E} \parallel \vec{B}$ and initial velocity is perpendicular to $\vec{E}$	R	Circle
D	$\vec{E} \neq \vec{B}, B = 0, \vec{E}$ perpendicular $\vec{B}$ and initial velocity is perpendicular to both $\vec{E}$ and $\vec{B}$	S	Helical path with non-uniform pitch
		T	Helical path with uniform pitch

- (1) (A)-P,R,T; (B)-P, Q; (C)-S; (D)-P
 - (2) (A)-P,T; (B)-P, Q; (C)-S; (D)-P
 - (3) (A)-P; (B)-P, Q; (C)-P; (D)-S
 - (4) (A)-P,T; (B)-P; (C)-S; (D)-P
42. A particle of specific charge (charge/mass) α starts moving from the origin under the action of an electric field $\vec{E} = E_0 \hat{i}$ and magnetic field $\vec{B} = B_0 \hat{k}$. Its velocity at $(x_0, y_0, 0)$ is $(4\hat{i} - 3\hat{j})$. The value of x_0 is

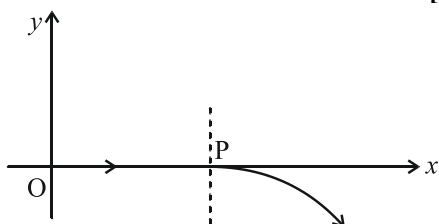
- (1) $\frac{13}{2} \frac{\alpha E_0}{B_0}$
- (2) $\frac{16 \alpha B_0}{E_0}$
- (3) $\frac{25}{2 \alpha E_0}$
- (4) $\frac{5 \alpha}{2 B_0}$

43.

Column-I (Magnetic moment of)		Column-II	
(A)	a uniformly charged ring rotating uniformly about its axis	(p)	$\frac{q\omega r^2}{5}$
(B)	a charged particle rotating uniformly about a point	(q)	$\frac{q\omega r^2}{4}$
(C)	a uniformly charged disk rotating uniformly about its axis	(r)	$\frac{q\omega r^2}{3}$
(D)	a uniformly charged spherical shell rotating uniformly about one of its diameter	(s)	$\frac{q\omega r^2}{2}$
(E)	a uniformly charged sphere rotating	(t)	$q\omega r^2$

44. For a positively charged particle moving in a x - y plane initially along the x -axis, there is a sudden change in its path due to the presence of electric and/or magnetic field beyond P. The curved path is shown in the x - y plane and is found to be non-circular. Which one of the following combinations is possible?

[JEE 2004]



45. An electron makes 3×10^5 revolutions per second in a circle of radius 0.5 angstrom. Find the magnetic field B at the centre of the circle.

- (1) 6×10^{-10} T (2) 12×10^{-10} T
(3) 18×10^{-10} T (4) 24×10^{-10} T

46. Two moving coil meters. M_1 and M_2 have the following particulars:

$$R_1 = 10 \Omega, N_1 = 30,$$

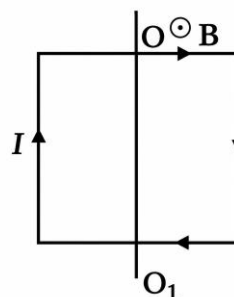
$$A_1 = 3.6 \times 10^{-3} \text{ m}^2, B_1 = 0.25 \text{ T}$$

$$R_2 = 14 \Omega, N_2 = 42, A_2 = 1.8 \times 10^{-3} \text{ m}^2, B_2 = 0.50 \text{ T}$$

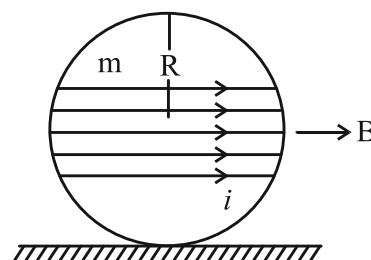
(The spring constants are identical for the two meters). Determine the ratio of

- (a) current sensitivity and
(b) voltage sensitivity of M_2 and M_1 .

47. A square current carrying loop made of thin wire and having a mass $m = 10\text{g}$ can rotate without friction with respect to the vertical axis OO_1 , passing through the centre of the loop at right angles to two opposite sides of the loop. The loop is placed in a homogeneous magnetic field with an induction $B = 10^{-1} \text{ T}$ directed at right angles to the plane of the drawing. A current $I = 2\text{A}$ is flowing in the loop. Find the period of small oscillations that the loop performs about its position of stable equilibrium.



48. A wire is wrapped $N = 10$ times over a solid sphere of mass $m = 5\text{kg}$, current $I = 2\text{A}$, which is placed on a smooth horizontal surface. A horizontal magnetic field of induction $|\vec{B}| = 10\text{T}$ is present. Find the angular acceleration experienced by the sphere. Assume that the mass of the wire is negligible compared to the mass of the sphere. If answer is $20n\pi$. Write value of n .

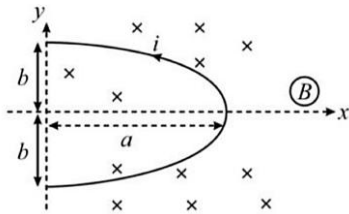


49. A particle of charge q and mass m starts moving from the origin under the action of an electric field $\vec{E} = E_0 \hat{i}$ and magnetic field $\vec{B} = B_0 \hat{i}$ with velocity $\vec{v} = v_0 \hat{j}$. The speed of the particle will become $2v_0$ after a time:

$$(1) \quad t = \frac{2mv_0}{qE} \quad (2) \quad t = \frac{2Bq}{mv_0}$$

$$(3) \quad t = \frac{\sqrt{3}Bq}{mv_0} \quad (4) \quad t = \frac{\sqrt{3}mv_0}{qE}$$

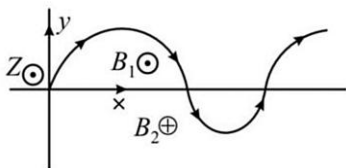
50. In the figure, there is a conducting wire having current i and which has a shape of half ellipse $\left[\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \right]$ is kept in a uniform magnetic field B as shown. If the mass of wire is m , the acceleration of wire will be



$$(1) \quad \frac{ibB}{m} \quad (2) \quad \frac{iaB}{m}$$

$$(3) \quad \frac{2iaB}{m} \quad (4) \quad \frac{2ibB}{m}$$

51. At $t = 0$ a charge q is at the origin and moving in the y -direction with velocity $\vec{v} = v \hat{j}$. The charge moves in a magnetic field that is for $y > 0$ out of page and given by $B_1 \hat{z}$ and for $y < 0$ into the page and given $-B_2 \hat{z}$. The charge's subsequent trajectory is shown in the sketch. From this information, we can deduce that

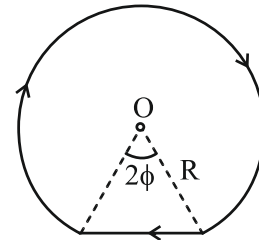


- (1) $q > 0$ and $|B_1| < |B_2|$
- (2) $q < 0$ and $|B_1| < |B_2|$
- (3) $q > 0$ and $|B_1| > |B_2|$
- (4) $q < 0$ and $|B_1| > |B_2|$

52. A current I flows along a thin wire shaped as a regular polygon with n sides which can be inscribed into a circle of radius R . Find the magnetic induction at the centre of the polygon. Analyse the obtained expression at $n \rightarrow \infty$.

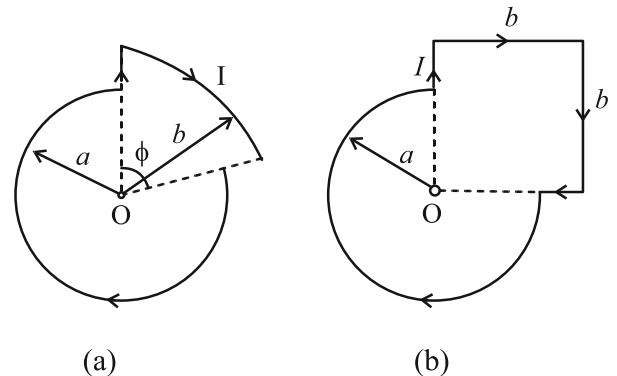
53. Find the magnetic induction at the centre of a rectangular wire frame whose diagonal is equal to $d = 16$ cm and the angle between the diagonals is equal to $\phi = 30^\circ$; the current flowing in the frame equals $I = 5.0$ A.

54. A current $I = 5.0$ A flows along a thin wire shaped as shown in Figure. The radius of a curved part of the wire is equal to $R = 120$ mm, the angle $2\phi = 90^\circ$. Find the magnetic induction of the field at the point O .

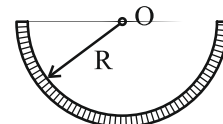


55. Find the magnetic induction of the field at the point O of a loop with current I , whose shape is illustrated.

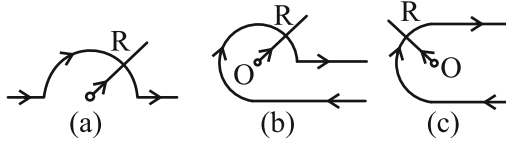
- (a) In Fig. a, the radii a and b , as well as the angle ϕ are known;
- (b) in Fig. b, the radius a and the side b are known.



56. A current I flows in a long straight wire with cross-section having the form of a thin half-ring of radius R (Figure). Find the induction of the magnetic field at the point O .

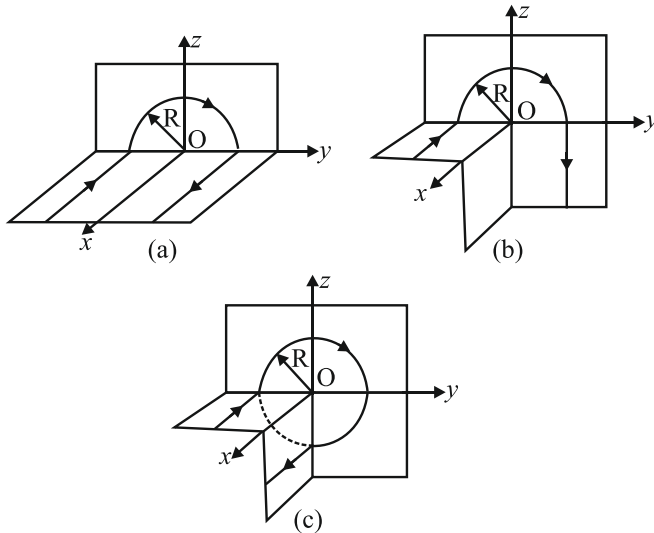


57. Find the magnetic induction of the field at the point O if a current-carrying wire has the shape shown in Figure (a), (b), (c). The radius of the curved part of the wire is R , the linear parts are assumed to be very long.



58. A very long wire carrying a current $I = 5.0$ A is bent at right angles. Find the magnetic induction at a point lying on a perpendicular to the wire, drawn through the point of bending, at a distance $l = 35$ cm from it.

59. Find the magnetic induction at the point O if the wire carrying a current $I = 8.0$ A has the shape shown in Figure (a), (b), (c). The radius of the curved part of the wire is $R = 100$ mm, the linear parts of the wire are very long



60. A current I flows along a round loop. Find the integral $\int B dr$ along the axis of the loop within the range from $-\infty$ to $+\infty$. Explain the result obtained.
61. A direct current of density j flows along a round uniform straight wire with cross-section radius R . Find the magnetic induction vector of this current at the point whose position relative to the axis of the wire is defined by a radius vector r . The magnetic permeability is assumed to be equal to unity throughout all the space.

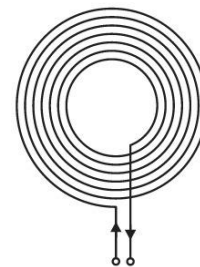
62. Inside a long straight uniform wire of round cross-section there is a long round cylindrical cavity whose axis is parallel to the axis of the wire and displaced from the latter by a distance l . A direct current of density j flows along the wire. Find the magnetic induction inside the cavity. Consider, in particular, the case $l = 0$.

63. A uniform current of density j flows inside an infinite plate of thickness $2d$ parallel to its surface. Find the magnetic induction induced by this current as a function of the distance x from the median plane of the plate. The magnetic permeability is assumed to be equal to unity both inside and outside the plate.

64. Find the magnetic moment of a thin round loop with current if the radius of the loop is equal to $R = 100$ mm and the magnetic induction at its centre is equal to $B = 6.0 \times T$.

65. A thin insulated wire forms a plane spiral of $N = 100$ tight turns carrying a current $I = 8$ mA. The radii of inside and outside turns (Fig) are equal to $a = 50$ mm and $b = 100$ mm. Find:

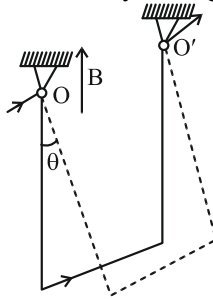
- (a) the magnetic induction at the centre of the spiral;
(b) the magnetic moment of the spiral with a given current.



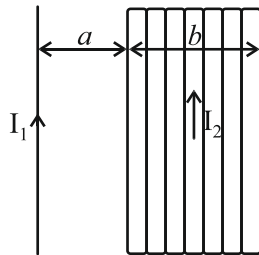
66. A non-conducting thin disc of radius R charged uniformly over one side with surface density σ rotates about its axis with an angular velocity ω . Find:
(a) the magnetic induction at the centre of the disc;
(b) the magnetic moment of the disc.

67. Find the magnetic moment of a thin round loop with current if the radius of the loop is equal to $R = 100$ mm and the magnetic induction at its centre is equal to $B = 6.0 \times T$.

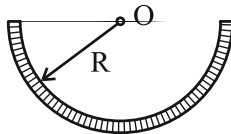
68. A copper wire with cross-sectional area $S = 2.5 \text{ mm}^2$ bent to make three sides a square can turn about a horizontal axis OO' (Fig.). The wire is located in uniform vertical magnetic field. Find the magnetic induction if on passing a current $I = 16 \text{ A}$ through the wire the latter deflects by an angle $\theta = 20^\circ$.



69. Two long thin parallel conductors of the shape shown in Figure carry direct currents I_1 and I_2 . The separation between the conductors is a , the width of the right-hand conductor is equal to b . With both conductors lying in one plane, find the magnetic interaction force between them reduced to a unit of their length.

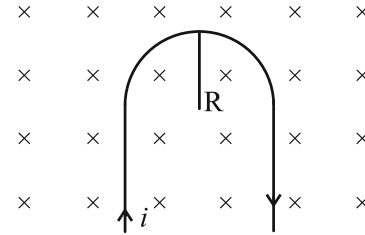


70. A direct current I flows in a long straight conductor whose cross-section has the form of a thin half-ring of radius R . The same current flows in the opposite direction along a thin conductor located on the "axis" of the first conductor (point O in Fig.). Find the magnetic interaction force between the given conductor reduced to a unit of their length.



71. A wire, carrying a current i , is kept in the x - y plane along the curve $y = A \sin\left(\frac{2\pi}{\lambda}x\right)$. A magnetic field B exists in the z -direction. Find the magnitude of the magnetic force on the portion of the wire between $x = 0$ and $x = \lambda$.

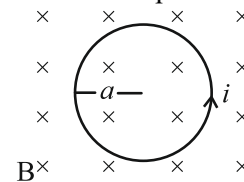
72. A rigid wire consists of a semicircular portion of radius R and two straight sections (figure). The wire is partially immersed in a perpendicular magnetic field B as shown in the figure. Find the magnetic force on the wire if it carries a current i .



73. A straight horizontal wire of mass 10 mg and length 1.0 m carries a current of 2.0 A . What minimum magnetic field B should be applied in the region so that the magnetic force on the wire may balance its weight?
74. A straight wire of length l can slide on two parallel plastic rails kept in a horizontal plane with a separation d . The coefficient of friction between the wire and the rails is μ . If the wire carries a current i , what minimum magnetic field should exist in the space in order to slide the wire on the rails.
75. Figure shows a circular wire-loop of radius a , carrying a current i , placed in a perpendicular magnetic field B .

- (a) Consider a small part dl of the wire. Find the force on this part of the wire exerted by the magnetic field.

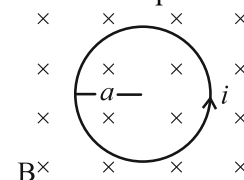
- (b) Find the force of compression in the wire.



76. Figure shows a circular wire-loop of radius a , carrying a current i , placed in a perpendicular magnetic field B .

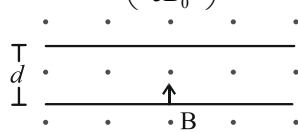
- (a) Consider a small part dl of the wire. Find the force on this part of the wire exerted by the magnetic field.

- (b) Find the force of compression in the wire.



77. Suppose that the radius of cross section of the wire used in the previous problem is r . Find the increase in the radius of the loop if the magnetic field is switched off. The Young modulus of the material of the wire is Y .
78. A particle having mass m and charge q is released from the origin in a region in which electric field and magnetic - field are given by

$$\vec{B} = -B_0 \vec{j} \text{ and } \vec{E} = E_0 \vec{k}.$$
Find the speed of the particle as a function of its z -coordinate.
79. An electron is emitted with negligible speed from the negative plate of a parallel plate capacitor charged to a - potential difference V . The separation between the plates is d and a magnetic field B exists in the space as shown in figure. Show that the electron will fail to strike the upper plate if

$$d > \left(\frac{2m_e V}{eB_0^2} \right)^{\frac{1}{2}}$$


80. A rectangular coil of 100 turns has length 5 cm and width 4 cm. It is placed with its plane parallel to a uniform magnetic field and a current of 2 A is sent through the coil. Find the magnitude of the magnetic field B , if the torque acting on the coil is 0.2 Nm^{-1} .
81. A 50-turn circular coil of radius 2.0 cm carrying a current of 5.0 A is rotated in a magnetic field of strength 0.20 T. (a) What is the maximum torque that acts on the coil? (b) In a particular position of the coil, the torque acting on it is half of this maximum. What is the angle between the magnetic field and the plane of the coil?



Answer Key

1. (1)
2. (2)
3. (3)
4. (3)
5. (1, 2, 3)
6. (3)
7. (2, 4)
8. (4)
9. (1, 3, 4)
10. (3, 4)
11. (1, 3)
12. (1, 2)
13. (7)
14. (2)
15. (1)
16. (4)
17. (5)
18. (1)
19. (3)
20. (4)
21. (4)
22. (1)
23. (1)
24. (2)
25. (2{1.99, 2.01})
26. (2)
27. (3)
28. (2)
29. (1, 4)
30. (1, 2, 4)
31. (1, 2)
32. (2)
33. (1, 2, 4)
34. (2)
35. (1, 3)
36. (3)
37. (1, 3)
38. (1)
39. (1, 3, 4)
40. (4)
41. (1)
42. (3)
43. ($A - s$, $B - s$, $C - q$, $D - r$, $E - p$)
44. (2)
45. (1)
46. (a) 1.4, (b) 1
47. $T_0 = 2\pi\sqrt{\frac{m}{6IB}} = 0.57\text{ s}$
48. (5)
49. (4)
50. (*)
51. (*)
52. $B = n \propto_0 I \tan(\pi n)/2\pi R$, for $n \rightarrow \infty$ $B = \propto_0 I/2R$
53. $B = 4\propto_0 I/\pi d \sin \phi = 0.10\text{ mT}$
54. $B = (\pi - \phi + \tan \phi) \propto_0 I/2\pi R = 28 \propto T$
55. (a) $B = \frac{\propto_0 I}{4\pi} \left(\frac{2\pi - \phi}{a} + \frac{\phi}{b} \right)$;
(b) $B = \frac{\propto_0 I}{4\pi} \left(\frac{3\pi}{4a} + \frac{\sqrt{2}}{b} \right)$
56. $B = \propto_0 I/\pi^2 R$
57. (a) $B = (\propto_0/4\pi) (\pi I/R)$;
(b) $B = (\propto_0/4\pi) (1 + 3\pi/2) I/R$
(c) $B = (\propto_0/4\pi) (2 + \pi) I/R$
58. $B = (\propto_0/4\pi) I \sqrt{2} / l = 2.0 \propto T$
59. (a) $B = (\propto_0/4\pi) \sqrt{4 + \pi^2} I/R = 0.30 \propto T$

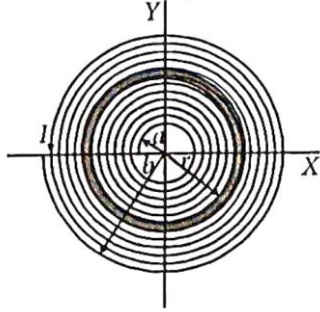


- (b) $B = (\mu_0/4\pi) \times \sqrt{2+2\pi+\pi^2} I/R = 0.34 \mu\text{T}$
- (c) $B = (\mu_0/4\pi) \sqrt{2} I/R = 0.11 \mu\text{T}$
60. The given integral is equal to $\mu_0 I$.
61. $B = \begin{cases} 1/2 \mu_0 [jr] & \text{for } r \leq R \\ 1/2 \mu_0 [jr] R^2 / r^2 & \text{for } r \geq R \end{cases}$
62. $B = 1/2 \mu_0 [jI]$, i.e., field inside the cavity is uniform
63. $B = \begin{cases} \mu_0 jx & \text{inside the plate.} \\ \mu_0 jd & \text{out side the plate.} \end{cases}$
64. $p_m = 2\pi R^2 B/\mu_0 = 30 \text{ mA}\cdot\text{m}^2$
65. (a) $B = \frac{\mu_0 IN \ln(b/a)}{2(b-a)} = 7 \mu\text{T}$
- (b) $p_m = 1/3 \pi IN(a^2 + ab + b^2) = 15 \text{ mA}\cdot\text{m}^2$
66. (a) $B = 1/2 \mu_0 \sigma \omega P$
- (b) $p_m = 1/4 \pi \sigma \omega R^4$
67. $p_m = 2\pi R^2 B/\mu_0 = 30 \text{ mA}\cdot\text{m}^2$.
68. $B = (2\rho g S/I) \tan\theta = mT$, where ρ is the density of copper.

69. $F_1 = \frac{\mu_0}{4\pi} \frac{2I_1 I_2}{b} \ln(1+b/a)$
70. $F_1 = \mu_0 I^2/\pi^2 R$
71. $(i\lambda B)$
72. $2iRB$, upward in the figure
73. $(4.9 \times 10^{-5} \text{ T})$
74. $\frac{\mu mg}{il}$
75. (a) id/B towards the centre, (b) iab
76. (a) id/B towards the centre, (b) iab
77. $\frac{ia^2 B}{\pi r^2 Y}$
78. $\sqrt{\frac{2qE_0 z}{m}}$
79. $(*)$
80. (0.5 T)
81. (a) $6.3 \times 10^{-2} \text{ N m}$, (b) 60°

Solution

1. (1)



Take an element of thickness dr at a distance r from the centre of spiral coil.

Number of turns in spiral = N

Number of turns per unit thickness

$$= \frac{N}{b-a}$$

Number of turns in element dr is, $dN = \frac{Ndr}{b-a}$

Magnetic field at the centre of the spiral due to current in element dr is

$$dB = \frac{\mu_0}{4\pi} \frac{2\pi dN I}{r} = \frac{\mu_0 I}{2r} dN = \frac{\mu_0 I}{2r} \left(\frac{Ndr}{b-a} \right)$$

$$= \frac{\mu_0 I}{2} \times \frac{N}{(b-a)} \times \frac{dr}{r}$$

Total magnetic field at the centre of the spiral due to current through the wire is

$$B = \int_a^b \frac{\mu_0 I N dr}{2(b-a)r} = \frac{\mu_0 I N}{2(b-a)} \int_a^b \frac{dr}{r} = \frac{\mu_0 I N}{2(b-a)} \ln\left(\frac{b}{a}\right)$$

2. (2)

Area of the loop $\hat{k}$

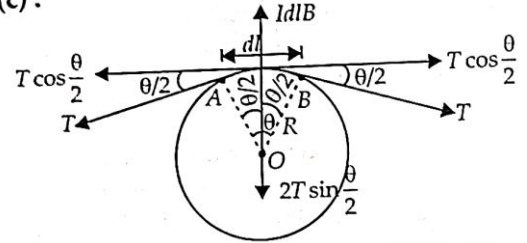
$$A = \left[a^2 + 4 \times \frac{\pi \left(\frac{a}{2} \right)^2}{2} \right] \hat{k} = \left[a^2 + \frac{\pi a^2}{2} \right] \hat{k}$$

Therefore, the magnetic moment of the current loop is

$$M = I \times A = I \left[a^2 + \frac{\pi a^2}{2} \right] \hat{k} = \left[1 + \frac{\pi}{2} \right] I a^2 \hat{k}$$

3. (3)

(c):



Consider an small element AB of length dl of the circle of radius R subtending an angle θ at the centre O .

If T is the tension in the wire, then force towards the centre will be equal to $2T \sin\left(\frac{\theta}{2}\right)$ which is balanced

by outward magnetic force on the current carrying element ($= IdlB$)

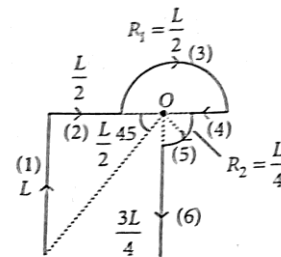
$$2T \sin\left(\frac{\theta}{2}\right) = IdlB$$

For small angle θ , $\sin\frac{\theta}{2} \approx \frac{\theta}{2}$

$$\text{or } T = \frac{IBdl}{\theta} = IBR \quad \left(\because \theta = \frac{dl}{R} \right)$$

$$= \frac{IBL}{2\pi} \quad \left(\because R = \frac{L}{2\pi} \right)$$

4. (3)



$$B_1 = \frac{\mu_0 I}{4\pi L} (\sin 0^\circ + \sin 45^\circ)$$

$$\vec{B}_1 = \frac{\mu_0 I}{4\sqrt{2}\pi L} (-\hat{k})$$

$$\vec{B}_2 = 0$$

$$\vec{B}_3 = \frac{\mu_0 I}{4\left(\frac{L}{2}\right)}(-\hat{k}) = \frac{\mu_0 I}{2L}(-\hat{k})$$

$$\vec{B}_4 = 0$$

$$\vec{B}_5 = \frac{\mu_0 I}{8\left(\frac{L}{4}\right)}(-\hat{k}) = \frac{\mu_0 I}{2L}(-\hat{k}); \vec{B}_6 = 0$$

$$\vec{B}_{\text{net}} = \vec{B}_1 + \vec{B}_2 + \vec{B}_3 + \vec{B}_4 + \vec{B}_5 + \vec{B}_6$$

$$\vec{B}_{\text{net}} = \frac{\mu_0 I}{L} \left(1 + \frac{1}{4\sqrt{2}\pi}\right)(-\hat{k})$$

$$= \frac{\mu_0 I}{2\pi} \left[\frac{1}{X_1} - \frac{1}{(X_0 - X_1)} \right]$$

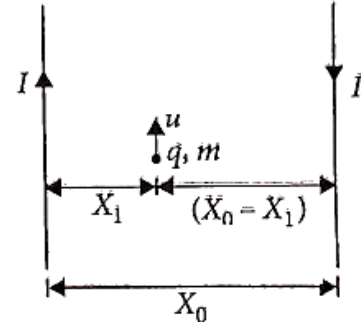
$$= \frac{\mu_0 I}{2\pi} \left[\frac{X_0 - X_1 - X_1}{X_1(X_0 - X_1)} \right]$$

$$= \frac{\mu_0 I}{2\pi} \left[\frac{X_0 - 2X_1}{X_1(X_0 - X_1)} \right]$$

$$R_1 = \frac{mu}{qB_1}$$

Case-2

When the currents are in the opposite direction



$$B_2 = \frac{\mu_0 I}{2\pi X_1} + \frac{\mu_0 I}{2\pi(X_0 - X_1)}$$

$$= \frac{\mu_0 I}{2\pi} \left[\frac{1}{X_1} + \frac{1}{(X_0 - X_1)} \right]$$

$$= \frac{\mu_0 I}{2\pi} \left[\frac{X_0 - X_1 + X_1}{X_1(X_0 - X_1)} \right]$$

$$B_2 = \frac{\mu_0 I}{2\pi} \left[\frac{X_0}{X_1(X_0 - X_1)} \right]$$

$$R_2 = \frac{mu}{qB_2}$$

$$\therefore \frac{R_1}{R_2} = \frac{B_1}{B_2} = \frac{X_0}{X_0 - 2X_1} = \frac{\frac{X_0}{X_1}}{\frac{X_0}{X_1} - 2} = \frac{3}{3-2} = 3$$

5. (1, 2, 3)

Magnetic force acting on a current carrying wire, kept in a uniform magnetic field,

$$\vec{F} = I(\vec{l} \times \vec{B})$$

Here, $\vec{l}$ = displacement of the wire = $2(L + R)\hat{x}$

$$\therefore \vec{F} = 2I(L + R)(\hat{x} \times \vec{B})$$

If $\vec{B} = B\hat{x}$ then

$$\vec{F} = 2I(L + R)(\hat{x} \times \hat{y})B = 0$$

If $\vec{B} = B\hat{y}$ then

$$\vec{F} = 2I(L + R)(\hat{x} \times \hat{x})B = 2IB(L + R)\hat{z} \text{ or } F \propto (L + R)$$

If $\vec{B} = B\hat{z}$ then

$$\vec{F} = 2I(L + R)(\hat{x} \times \hat{z})B = -2IB(L + R)\hat{y}$$

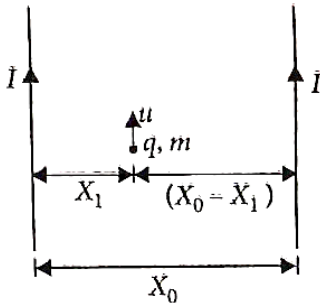
or, $F \propto (L + R)$

So, options (1), (2) and (3) are correct.

6. (3)

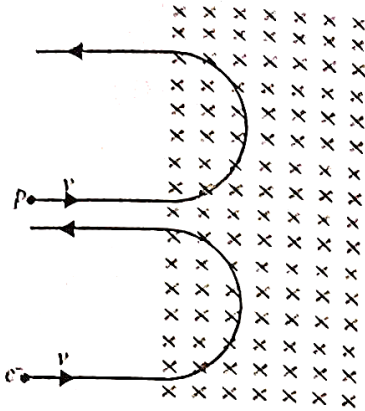
Case-1

When the currents are in the same direction



$$B_1 = \frac{\mu_0 I}{2\pi X_1} - \frac{\mu_0 I}{2\pi(X_0 - X_1)}$$

7. (2, 4)



From figure, it is clear that they will come out travelling along parallel paths. Hence, (2) is correct. The velocity of the charged particle is normal to the magnetic field and the magnetic field is uniform, so the particle will follow a circular path. The radius of the circular path is

$$R = \frac{mv}{qB}$$

For proton, $R_p = \frac{m_p v}{eB}$; For electron, $R_e = \frac{m_e v}{eB}$
 $m_p > m_e \Rightarrow R_p > R_e$

Time spent by the particle in the field is $T = \frac{\pi m}{qB}$

$\therefore$ For electron, $T_e = \frac{\pi m_e}{eB}$; For proton, $T_p = \frac{\pi m_p}{eB}$

$m_p > m_e \Rightarrow T_p > T_e$

Hence, (4) is correct

8. (4)

Mass on α -particle, $m_\alpha = 4$ amu
 Mass on sulphur ion, $m_s = 32$ amu
 Potential = V

The KE is given by, $\frac{1}{2} m v^2 = qV$

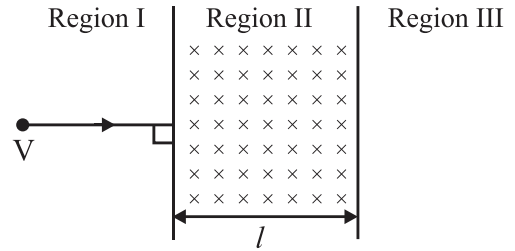
$$\Rightarrow v = \sqrt{\frac{2qV}{m}}$$

Let the radius is r .

$$r = \frac{mv}{Bq} = \sqrt{\frac{2mqV}{qB}} = \frac{1}{B} \sqrt{\frac{2mV}{q}}$$

$$\text{So, } \frac{r_s}{r_\alpha} = \sqrt{\frac{m_s}{m_\alpha} \cdot \frac{q_\alpha}{q_s}} = \sqrt{\frac{32}{4} \times \frac{2}{1}}; \frac{r_s}{r_\alpha} = 4$$

9. (1, 3, 4)



The velocity of the particle is normal to the magnetic field, and the magnetic field is uniform, so the particle will follow a circular path.

The radius of the circular path is

$$R = \frac{mV}{qB}$$

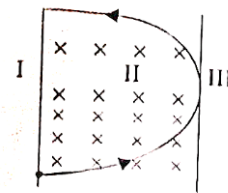
For the particle to enter region III

$$R > l \text{ or } \frac{mV}{qB} > l.$$



For path length of particle in region II to be maximum

$$l = R \text{ or } V = \frac{qB}{m}.$$



The period of revolution of charged particle is

$$\omega = \frac{qB}{m}.$$

The time spent in region II is

$$t = \frac{\pi}{\omega} = \frac{\pi m}{qB}, \text{ which is same for all the velocities for}$$

which the charged particle returns to region I.

10. (3, 4)

For a positive charge,

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

Given:

$$\vec{E} = E_0 \hat{j}, \vec{B} = B_0 \hat{j}$$

The electric field accelerates the charge along the y -axis, while the magnetic field causes circular motion in the $x - z$ plane.

(1) False: For $\theta = 0^\circ$, v_y starts from zero but increases due to E , so the motion is helical, not purely circular.

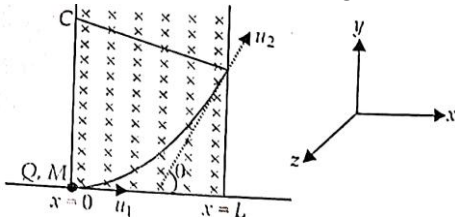
(2) False: Since v_y increases with time, the pitch is not constant.

(3) True: For $\theta = 10^\circ$, v_y increases with time, so the helical pitch increases along the y -axis.

(4) True: For $\theta = 90^\circ$, velocity is along y , so magnetic force is zero and the charge has linearly accelerated motion along y .

11. (1, 3)

The situation is as shown in the figure.



Here, $u_1 = 4\hat{i} \text{ ms}^{-1}$

$u_2 = 2(\sqrt{3}\hat{i} + \hat{j})\text{ms}^{-1}$

When the charged particle enters in an uniform magnetic field, it travels a circular path.

Component of final velocity of particle is in positive y -direction.

Centre of circle is present on positive y -axis.

So, magnetic field is present in negative z -direction.

Angle of deviation

$$\tan \theta = \frac{u_{2y}}{u_{2x}} = \frac{2}{2\sqrt{3}} = \frac{1}{\sqrt{3}}$$

$$\theta = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = 30^\circ = \frac{\pi}{6}$$

$$\omega = \frac{\theta}{t}$$

$$\therefore t = \frac{\theta}{\omega} = \frac{\theta}{(QB/M)} = \frac{M\theta}{QB} \quad \left(\because \omega = \frac{QB}{M}\right)$$

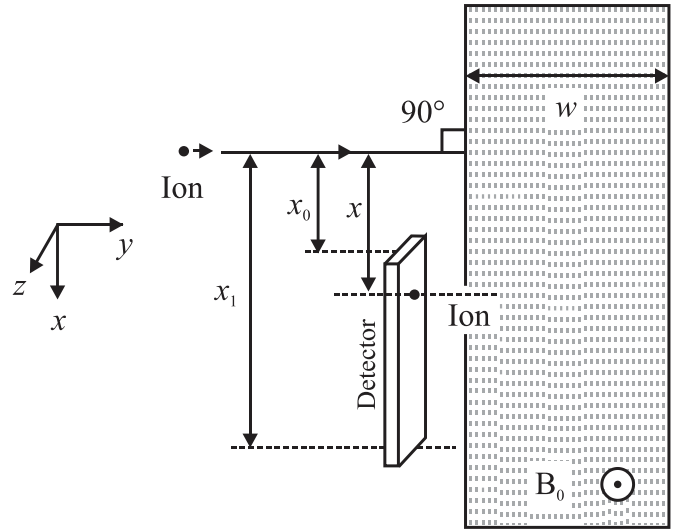
$$\therefore B = \frac{M\theta}{Qt} = \frac{M\pi}{6Q \times 10 \times 10^{-3}} \quad (\because t = 10 \text{ ms} = 10^{-3} \text{ s})$$

$$= \frac{100M\pi}{6Q} = \frac{50\pi M}{3Q}$$

12. (1, 2)

$$V = 192 \text{ V}, B_0 = 0.1 \hat{k} \text{ T}$$

$$m = 5/3 \times 10^{-27} \text{ kg}, q = 1.6 \times 10^{-19} \text{ C}$$



$$x = 2 \times \text{radius} = 2R = \frac{2mv}{Bq}$$

$$\therefore x = \frac{2}{qB} \sqrt{2m(q\Delta V)} = \frac{2}{B} \sqrt{\frac{2mV}{q}}$$

$$\text{For } H^+ \text{ ion, } x = \frac{2}{0.1} \sqrt{\frac{2 \times 5/3 \times 10^{-27} \times 192}{1.6 \times 10^{-19}}}$$

$$x = 4 \text{ cm}$$

So, option (1) is correct.

For $A_M = 144$:

$$x = 12 \times x_H^+$$

$$\therefore x = 12 \times 4 = 48 \text{ cm}$$

So, option (2) is correct.

For $1 \leq A_M \leq 196$

$$(x_1 - x_0)_{\min} = (\sqrt{A_{M196}})x_{H^+} - (\sqrt{A_{M1}})x_{H^+}$$

$$= 14 \times 4 - 4 = 52 \text{ cm}$$

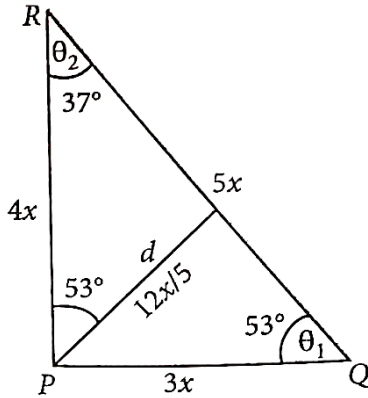
So, option (3) is incorrect.

For $A_M = 196$

$$\text{Width}_{\min} = \frac{R_{196}}{2} = 28 \text{ cm}$$

So, option (4) is incorrect.

13. (7)



B due to elements PR and PQ = 0 as the point P is on the line of the conductor. The magnetic induction due to the current element RQ at P.

$$B = \frac{\mu_0 I}{4\pi} \frac{(\sin \theta_1 + \sin \theta_2)}{d}$$

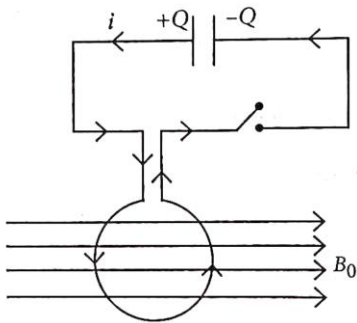
$$B = \frac{\mu_0 I}{4\pi(12x/5)} \left(\frac{4}{5} + \frac{3}{5} \right)$$

$$B = \left(\frac{\mu_0 I}{48\pi x} \right) 7 = \left(\frac{\mu_0 I}{48\pi x} \right) k \quad \therefore k = 7$$

14. (2)

Here, the net torque acting on the coil due to the external magnetic field is B_0

$$\vec{\tau} = |\vec{m} \times \vec{B}_0| = NiA \times B_0 = NiAB_0 \sin 90^\circ$$



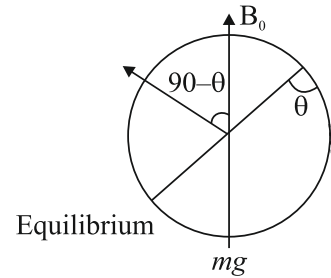
We know that the change in angular momentum,

$$\Delta L = \tau \Delta t = \int \tau dt = \int NiAB_0 dt$$

$$= NAB_0 \int i dt = NAB_0 Q \quad \left[\because \int i dt = Q \right]$$

$$\text{or } \Delta L = N\pi R^2 B_0 Q$$

15. (1)



$$\tau_0 = MB_0 \cos \theta \quad \dots(i)$$

$$\tau = mgr \sin \theta \quad \dots(ii)$$

Equate, eq (i) and (ii),

$$MB_0 \cos \theta = mgr \sin \theta$$

$$I\pi r^2 B_0 \cos \theta = mgr \sin \theta \quad (\because M = I\pi r^2)$$

$$\tan \theta = \frac{I\pi r B_0}{mg}$$

16. (4)

For an infinitely long conducting cylinder, by Ampere's law:

$$B(2\pi r) = \mu_0 I_{\text{enclosed}}$$

Let inner radius $a = R/2$.

- For $r < R/2$: no current is enclosed, so

$$B = 0$$

- For $R/2 < r < R$:

$$B = \frac{\mu_0 I}{2} \left(r - \frac{(R/2)^2}{r} \right)$$

so B increases from zero.

- For $r > R$:

$$B \propto \frac{1}{r}$$

so B decreases.

Hence the graph is zero up to $R/2$, rises up to R , and then falls as $1/r$.

Correct option: (4)

17. (5)

Magnetic field at point P due to cylinder

$$B_1 = \frac{\mu_0 J \pi a^2}{2\pi a} = \frac{\mu_0 J a}{2}$$

Magnetic field at point P due to cavity

$$B_2 = \frac{\mu_0}{2\pi} \frac{J\pi \left(\frac{a}{2} \right)^2}{\left(\frac{3a}{2} \right)} = \frac{\mu_0 J a}{12}$$



The magnetic field at point P is

$$B = B_1 - B_2 = \frac{\mu_0 J a}{2} - \frac{\mu_0 J a}{12}$$

$$= \frac{\mu_0 J a}{2} \left[1 - \frac{1}{6} \right] = \frac{5}{12} \mu_0 a J = \frac{N}{12} \mu_0 a J$$

$$\therefore N = 5$$

18. (1)

The force in the presence of electric and magnetic field is given as

$$\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})$$

For particle to move in straight line with constant velocity

$$\vec{E} + \vec{v} \times \vec{B} = 0 \quad \dots(i)$$

$$\therefore \vec{v} = \frac{E_0}{B_0} \hat{y} \text{ and } \vec{E} = -E_0 \hat{x}$$

$$\vec{B} = B_0 \hat{z}$$

This will be direction of electric and magnetic field to satisfy the eqn. (i).

So, option (1) is correct.

19. (3)

A particle will describe a helical path along positive z direction for

$$\vec{v} = 2 \frac{E_0}{B_0} \hat{x}$$

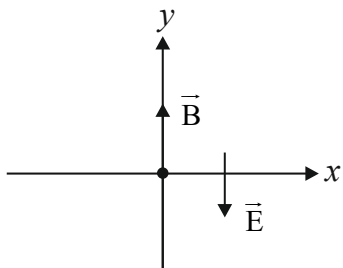
And magnetic field and electric field should be along z-direction.

$$\therefore \vec{E} = E_0 \hat{z}$$

$$\text{and } \vec{B} = B_0 \hat{z}$$

So option (3) is correct.

20. (4)



Particle will move in a straight line along the negative direction of y-axis, if electric force on it is along negative y-axis and magnetic force is zero.

21. (4)

$$[B] = [MT^{-2}I^{-1}]$$

$$[e] = [IT], [m_e] = [M], [h] = [ML^2T^{-1}]$$

$$[k] = [ML^3T^{-4}I^{-2}]$$

Given,

$$[B] = [e]^\alpha [m_e]^\beta [h]^\gamma [k]^\delta$$

Comparing powers:

$$\beta + \gamma + \delta = 1$$

$$2\gamma + 3\delta = 0$$

$$\alpha - \gamma - 4\delta = -2$$

$$\alpha - 2\delta = -1$$

Solving:

$$\alpha = 3, \beta = 2, \gamma = -3, \delta = 2$$

Therefore,

$$\alpha + \beta + \gamma + \delta = 3 + 2 - 3 + 2 = 4$$

22. (1)

Since the quantity is dimensionless, the net powers of M, L, T, I must be zero.

$$[e] = [IT]$$

$$[\epsilon_0] = [M^{-1}L^{-3}T^4I^2]$$

$$[h] = [ML^2T^{-1}], [c] = [LT^{-1}]$$

For

$$[e]^\alpha [\epsilon_0]^\beta [h]^\gamma [c]^\delta$$

comparing dimensions gives:

$$\alpha + 2\beta = 0$$

$$-\beta + \gamma = 0$$

$$-3\beta + 2\gamma + \delta = 0$$

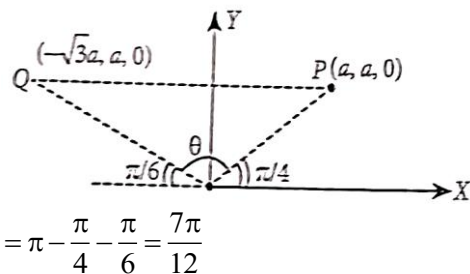
Solving:

$$\alpha = 2n, \beta = -n, \gamma = -n, \delta = -n$$

Therefore,

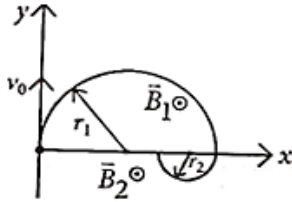
$$(\alpha, \beta, \gamma, \delta) = (2, -n, -n, -n)$$

23. (1)



$$\text{So, } \int \vec{B} \cdot d\vec{l} = \frac{\mu_0 I}{2\pi} (\theta) = \frac{\mu_0 I}{2\pi} \times \frac{7\pi}{12} = \frac{7\mu_0 I}{24}$$

24. (2)



$$\text{Average speed along x-axis, } v_x = \frac{d_1 + d_2}{t_1 + t_2}$$

$$\text{We have, } r_1 = \frac{mv_0}{qB_1} \text{ or } r_2 = \frac{mv_0}{qB_2}$$

$$\text{Since } B_1 = \frac{B_2}{4}$$

$$\therefore r_1 = 4r_2$$

$$\text{Time spent by charged particle in } B_1 \text{ is } t_1 = \frac{\pi m}{qB_1}$$

$$\text{Time spent by charged particle in } B_2 \text{ is } t_2 = \frac{\pi m}{qB_2}$$

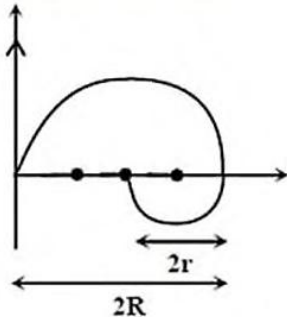
Total distance along x-axis is

$$d_1 + d_2 = 2r_1 + 2r_2 = 2(r_1 + r_2) = 2(5r_2) = 10r_2$$

$$\text{Average speed} = \frac{10r_2}{5t_2} = 2 \frac{mv_0}{qB_2} \times \frac{qB_2}{\pi m} = 2 \text{ ms}^{-1}$$

$$(\because v_0 = \pi \text{ ms}^{-1})$$

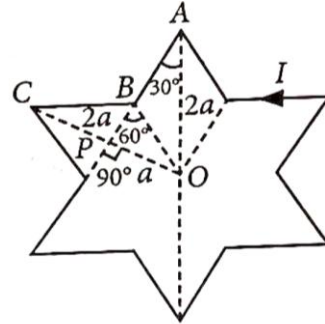
25. (2{1.99, 2.01})



Avg. speed along x-axis

$$\begin{aligned} &= \frac{\text{total distance travelled along x-axis}}{\text{total time taken}} \\ &= \frac{2R + 2r}{\frac{\pi R}{V_0} + \frac{\pi r}{V_0}} = \frac{2V_0}{\pi} = 2 \text{ m/s} \end{aligned}$$

26. (2)



Let us first calculate the magnetic field due to element AB.

Perpendicular distance of centre O from element AB, $OP = a$

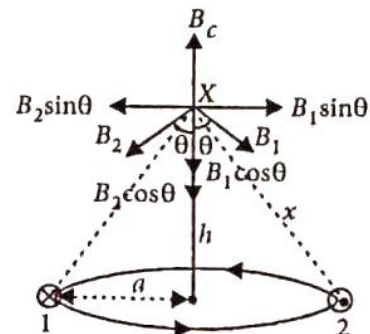
Angle subtended by element at centre are $\theta_1 = 30^\circ$, $\theta_2 = 60^\circ$ as shown in figure.

$$\begin{aligned} \therefore \vec{B} &= \frac{\mu_0 I}{4\pi a} (\cos 30^\circ - \cos 60^\circ) \odot \\ &= \frac{\mu_0 I}{4\pi a} \left(\frac{\sqrt{3}}{2} - \frac{1}{2} \right) \odot = \frac{\mu_0 I}{8\pi a} (\sqrt{3} - 1) \odot \end{aligned}$$

Since the direction of magnetic field due to each element side is out of the paper

$$\begin{aligned} \therefore \text{Net magnetic field at O} &= 12 \times \frac{\mu_0 I}{8\pi a} (\sqrt{3} - 1) \\ &= \frac{\mu_0 I}{4\pi a} 6(\sqrt{3} - 1) \end{aligned}$$

27. (3)



Let magnetic field due to circular loop = $\vec{B}_c$

Magnetic field due to wire = 1 = $\vec{B}_1$

Magnetic field due to wire = 2 = $\vec{B}_2$

In magnitude, $B_1 = B_2 = \frac{\mu_0 I}{2\pi x}$

So, $B_1 \sin \theta$ and $B_2 \sin \theta$ cancelled each other.

Resultant of B_1 and $B_2 = B_2 \cos \theta + B_2 \cos \theta$

$$= 2B_1 \cos \theta = \frac{2\mu_0 I}{2\pi x} \times \frac{h}{x} = \frac{\mu_0 I h}{\pi x^2}$$

(From figure, $\cos \theta = h/x$)

$$= \frac{\mu_0 I h}{\pi(a^2 + h^2)}$$

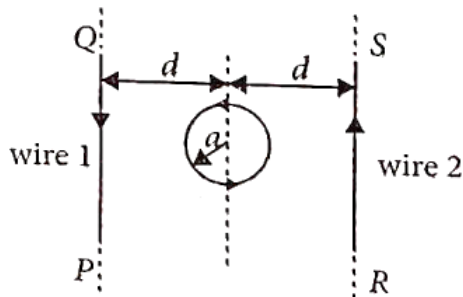
$$B_c = \frac{\mu_0 I a^2}{2(a^2 + h^2)^{3/2}}$$

For zero magnetic field at X,

$$\frac{\mu_0 I h}{\pi(a^2 + h^2)} = \frac{\mu_0 I a^2}{2(a^2 + h^2)^{3/2}}$$

$$\Rightarrow \frac{h}{\pi} = \frac{a^2}{2\sqrt{(a^2 + h^2)}} \Rightarrow h \approx 1.2a$$

28. (2)



The magnitude of torque on the loop is given by

$$\tau = MB \sin \theta$$

Here, $M = IA = I(\pi a^2)$

$$B = 2 \times \frac{\mu_0 I}{2\pi d}$$

(net magnetic field at the mid point of the loop) $\theta = 30^\circ$

$$\tau = I(\pi a^2) \times \frac{\mu_0 I}{\pi d} \times \sin 30^\circ = \frac{\mu_0 I^2 a^2}{2d}$$

29. (1, 4)

There are two magnetic fields:

1. Infinite solenoid:

$$B_s = \mu_0 n I$$

inside $r < 2R$, directed along the axis.

2. Cylindrical

conductor:

For $r > R$,

$$B_c = \frac{\mu_0 I}{2\pi r}$$

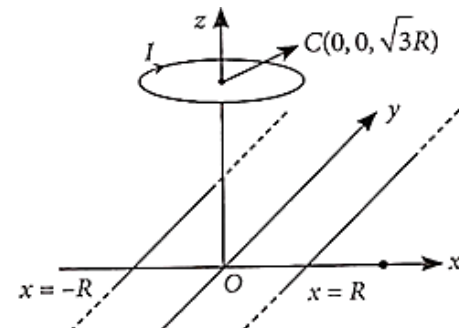
directed tangentially.

Now check the options:

- (1) True: For $0 < r < R$, solenoid field is non-zero.
- (2) False: For $R < r < 2R$, both axial and tangential fields exist, so resultant is not purely axial.
- (3) False: Resultant is not purely tangential.
- (4) True: For $r > 2R$, solenoid field is zero, but conductor produces non-zero tangential field.

Correct options: (1) and (4)

30. (1, 2, 4)



Magnetic field due to ring at origin

$$= \frac{\mu_0 \times I \times R^2}{2 \times 8R^3} \left(-\hat{k} \right) = \frac{\mu_0 I}{16R} \left(-\hat{k} \right)$$

Magnetic field due to ring at origin due to wires

$$\vec{B}_1 + \vec{B}_2 = \left(\frac{\mu_0 I_1}{2\pi R} - \frac{\mu_0 I_2}{2\pi R} \right) \hat{k}$$

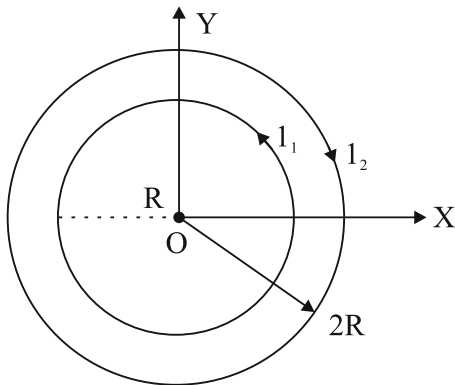
(Assume current I_1 and I_2 are along $+\hat{j}$ direction.)

(a) If $I_1 = I_2$, then $\vec{B}_O = \frac{\mu_0 I}{16R} \left(-\hat{k} \right)$. It cannot be zero.

(b) If $I_1 > 0$ and $I_2 < 0$, $\vec{B}$ at origin due to wires will be along $+\hat{k}$ direction and $\vec{B}$ due to ring is along $-\hat{k}$ direction. Hence net magnetic field at point O can be equal to zero.

- (c) If $I_1 > 0, I_2 > 0$, then net magnetic field at origin will be $\vec{B}_O = \left[\frac{\mu_0 (I_1 + I_2)}{2\pi R} + \frac{\mu_0 \cdot I}{16R} \right] \hat{k}$. It cannot be zero.
- (d) As $I_1 = I_2$, then $\vec{B}_1 = -\vec{B}_2$, so magnetic field along z-axis is only due to the ring. Magnetic field at centre of the loop is $\vec{B}_1 = \frac{\mu_0 I}{2R}$ in-z direction.

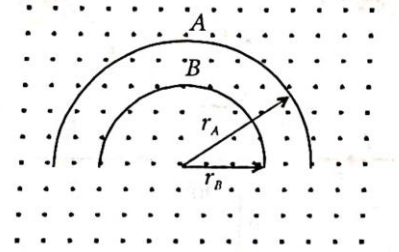
31. (1, 2)



- (a) From Biot Savart's law,
- $$d\vec{B} = \frac{\mu_0}{4\pi} \cdot \frac{id\vec{l}}{r^3}$$
- Here, $d\vec{l}$ is in x - y plane and $\vec{r}$ also in xy plane.
- (b) Due to symmetry, it depends on r .
- $$r = \sqrt{x^2 + y^2}$$
- (c) The magnetic field at centre due to current I_1 is
- $$B_1 = \frac{\mu_0 I_1}{2R_1} = \frac{\mu_0 I_1}{2R}$$
- And due to current I_2 is,
- $$B_2 = \frac{\mu_0 I_2}{2R_2} = \frac{\mu_0 I_2}{4R}$$
- So, net magnetic field is zero somewhere between the loops.
- (d) The net direction of magnetic field depends on the value of B_1 and B_2 .

32. (2)

Magnetic force on the charged particle provides centripetal force for circular motion of the charged particle, when the magnetic field exists perpendicular to the plane of motion.



For particle A, $\frac{m_A v_A^2}{r_A} = qv_A \times (\text{field } F)$

or $r_A = \frac{m_A v_A}{qF}$ For particle B, $r_B = \frac{m_B v_B}{qF}$

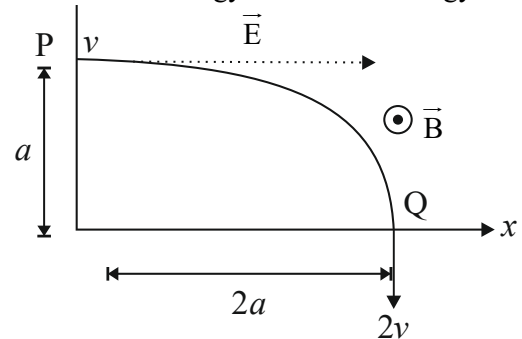
$$\therefore \frac{r_A}{r_B} = \frac{m_A v_A}{m_B v_B}$$

Since $r_A > r_B$ according to given figure, $m_A v_A > m_B v_B$.

33. (1, 2, 4)

- (a) Magnetic force does not perform any works as it is at right angle to direction of motion of the charged particle.

$\therefore$ Potential energy = Δ Kinetic energy



$$\text{or } (qE)(2a) = \frac{1}{2} m(4v^2 - v^2) \text{ or } E = \frac{3}{4} \left(\frac{mv^2}{qa} \right)$$

(1) is correct option.

(2) At P, rate of work done = $(qE)v$

$$= q \left(\frac{3}{4} \frac{mv^2}{qa} \right) v = \frac{3}{4} \left(\frac{mv^3}{a} \right)$$

Option (2) is correct and option (3) is incorrect.

(4) Rate of workdone at Q:

Magnetic field does no work.

By electric field, work is done.

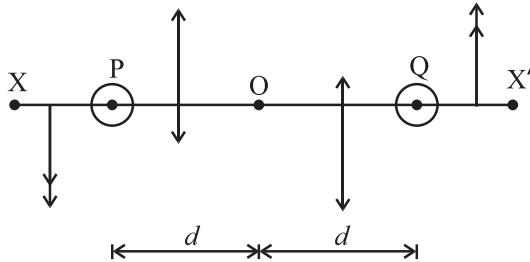
$$\therefore \text{Rate of workdone} = (qE) \times (2v) \times \cos 90^\circ = \text{zero}$$

$$\therefore \text{Rate of workdone by both the fields at Q is zero.}$$

34. (2)

The steady currents in the two parallel wires flow out of paper. O is a point in the centre between two wires. The wires are at P and Q. Since the current flows out of paper, the magnetic field at points to the right of given wire will be upwards and to the left will be downwards.

- (i) At point O, net field will be zero. The fields due to wires at P and Q are equal in magnitude but opposite in directions.



- (ii) Region OX': Magnetic field will be upwards. The region lies to right of both the wires.
 (iii) Region OQ: Net field will be downwards. Due to nearness, wire at Q dominates.
 (iv) Region OP: Net field will be upwards. Due to nearness, wire at P dominates.
 (v) Region PX: Net field will be downwards. The region lies to left of both the wires.

Graph (2) satisfies all the five conditions.

35. (1, 3)

Using the magnetic field due to the long wire,

$$B = \frac{\mu_0 I_1}{2\pi r}$$

The wire carries current into the paper, so the magnetic field is clockwise around O.

- On the two radial sides BC and AD, magnetic forces are equal and opposite, so

$$F_{\text{net}} = 0$$

Hence (1) is correct.

- These two opposite forces form a couple, producing a torque about OO'. The torque is clockwise when seen from O.

Therefore,

$$\text{Correct options: (1) and (3)}$$

36. (3)

For a proton moving with constant velocity,

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}) = 0$$

Thus, the electric and magnetic forces must cancel each other when both fields are present.

- (1) $E = 0, B = 0$: Possible $\rightarrow$ no force.
- (2) $E = 0, B \neq 0$: Possible only if $\vec{v} \parallel \vec{B}$.
- (3) $E \neq 0, B = 0$: Not possible $\rightarrow$ electric force $qE \neq 0$.
- (4) $E \neq 0, B \neq 0$: Possible if $\vec{E}$ and $\vec{v} \times \vec{B}$ cancel.

$$\text{Correct answer: (3)}$$

37. (1, 3)

Let the radius of circular motion be

$$r = \frac{P}{qB}$$

From the geometry of the path,

$$d = r \sin 30^\circ = \frac{r}{2}$$

Hence,

$$P = 2qBd$$

So (1) is correct.

For maximum change in momentum, the particle must complete a semicircular path. This is possible when

$$d \geq r = \frac{P}{qB}$$

Thus (3) is correct.

38. (1)

Since the charge moves without acceleration, net force is zero:

$$q(\vec{E} + \vec{v} \times \vec{B}) = 0$$

Therefore,

$$\vec{v} \times \vec{B} = -\vec{E}$$

Given:

$$E = 10^3 \hat{i}, B = 2 \times 10^{-2} \hat{j}$$

For $\vec{v} = v_z \hat{k}$,

$$\vec{v} \times \vec{B} = v_z (\hat{k} \times \hat{j}) B = -v_z B \hat{i}$$

Thus,

$$v_z B = 10^3$$

$$v_z = \frac{10^3}{2 \times 10^{-2}} = 5 \times 10^4 \text{ m/s}$$

Hence,

$$\vec{v} = 5 \times 10^4 \hat{k} \text{ m/s}$$

Correct option: (1)

39. (1, 3, 4)

For the original wire,

$$\vec{J} = -J_0 \hat{k}$$



The cavity can be treated as a cylinder carrying current $+J_0\hat{k}$. By superposition,

$$\vec{B}_{\text{cavity}} = \frac{\mu_0}{2}(\vec{J} \times \vec{a})$$

Since $\vec{a} = a\hat{i}$,

$$\vec{B}_{\text{cavity}} = \frac{\mu_0 J_0 a}{2} \hat{j}$$

Thus, the magnetic field inside the cavity is uniform and perpendicular to $\vec{a}$.

For the electron,

$$\vec{v} = v_0 \hat{j} \parallel \vec{B}$$

Therefore,

$$\vec{F} = q(\vec{v} \times \vec{B}) = 0$$

So it moves undeviated until it collides with the cavity wall.

Correct options: (1), (3), (4)

40. (4)

For an electron in a magnetic field, the circular frequency is

$$f = \frac{eB}{2\pi m_e}$$

For a long solenoid,

$$B = \mu_0 nI$$

Therefore,

$$f = \frac{e\mu_0 nI}{2\pi m_e}$$

So,

$$n = \frac{2\pi m_e f}{e\mu_0 I}$$

Substituting,

$$n = \frac{2\pi(9.1 \times 10^{-31})(10^8)}{(1.6 \times 10^{-19})(4\pi \times 10^{-7})(2)}$$

$$n \approx 1420 \text{ turns/m}$$

Correct option: (4) 1420 Turns/m

41. (1)

Using Lorentz force,

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

- A: $E = 0, B \neq 0$, velocity at any angle $\rightarrow$ circular or helical motion.

$$A \rightarrow P, R, T$$

- B: $E \neq 0, B = 0 \rightarrow$ constant acceleration $\rightarrow$ straight line or parabola.

$$B \rightarrow P, Q$$

- C: $E \parallel B, v \perp E \rightarrow$ circular motion due to B with increasing velocity along $E \rightarrow$ non-uniform pitch helix.

$$C \rightarrow S$$

- D: $E \perp B, v \perp E, B$. For $v = E/B$, electric and magnetic forces can cancel $\rightarrow$ straight line is possible.

$$D \rightarrow P$$

Therefore,

$$A \rightarrow P, R, T; B \rightarrow P, Q; C \rightarrow S; D \rightarrow P$$

42. (3)

Since the magnetic field does no work, only the electric field changes the particle's kinetic energy.

Starting from rest:

$$\frac{1}{2}mv^2 = qE_0x_0$$

Using specific charge,

$$\frac{1}{2}v^2 = \alpha E_0x_0$$

Given

$$\vec{v} = 4\hat{i} - 3\hat{j}$$

so

$$v^2 = 4^2 + (-3)^2 = 25$$

Therefore,

$$x_0 = \frac{25}{2\alpha E_0}$$

$$x_0 = \frac{25}{2\alpha E_0}$$

Correct option: (3)

43. (A - s, B - s, C - q, D - r, E - p)

For a rotating charged body,

$$\mu = \frac{q}{2M}I\omega$$

where I is the moment of inertia about the axis of rotation.

- (A) Ring: $I = Mr^2$

$$\mu = \frac{q\omega r^2}{2} \Rightarrow [S]$$

- (B) Charged particle: $I = Mr^2$

$$\mu = \frac{q\omega r^2}{2} \Rightarrow [S]$$

- (C) Disk: $I = \frac{1}{2}Mr^2$

$$\mu = \frac{q\omega r^2}{4} \Rightarrow [q]$$

- (D) Spherical shell: $I = \frac{2}{3}Mr^2$



$$\mu = \frac{q\omega r^2}{3} \Rightarrow [r]$$

- (E) Solid sphere: $I = \frac{2}{5}Mr^2$

$$\mu = \frac{q\omega r^2}{5} \Rightarrow [p]$$

$$[A - s, B - s, C - q, D - r, E - p]$$

44. (2)

If $\vec{E} = 0$, the path of the charged particle beyond P will be helix. It is not given. Option (1) and (c) are obviously incorrect.

Option (4) is incorrect because if $\vec{B}$ is in y - z plane, the net force on the particle will be in $\hat{k}$ direction. The motion beyond P is in x - y plane according to the question. The particle can move in x - y plane only in option (2). The path will be non-circular as the electric field $\vec{E}$ exerts force in $+x$ direction.

45. (1)

For an electron moving in a circular path, the current is

$$I = ef$$

Magnetic field at the centre:

$$B = \frac{\mu_0 I}{2r} = \frac{\mu_0 ef}{2r}$$

Given:

$$f = 3 \times 10^5 \text{ s}^{-1}, r = 0.5 \text{ Å} = 0.5 \times 10^{-10} \text{ m}$$

Therefore,

$$B = \frac{(4\pi \times 10^{-7})(1.6 \times 10^{-19})(3 \times 10^5)}{2(0.5 \times 10^{-10})}$$

$$[B \approx 6 \times 10^{-10} \text{ T}]$$

Correct option: (1)

46. (a) 1.4, (b) 1

Yes, your answer is correct. I made an error in the voltage-sensitivity ratio.

For a moving coil meter:

$$S_I = \frac{NBA}{k}$$

(a) Current sensitivity

$$\begin{aligned} \frac{S_{I2}}{S_{I1}} &= \frac{N_2 A_2 B_2}{N_1 A_1 B_1} \\ &= \frac{42(1.8 \times 10^{-3})(0.50)}{30(3.6 \times 10^{-3})(0.25)} = 1.4 \end{aligned}$$

$$[a] = 1.4$$

(b) Voltage sensitivity

$$S_V = \frac{S_I}{R}$$

Therefore,

$$\begin{aligned} \frac{S_{V2}}{S_{V1}} &= 1.4 \times \frac{R_1}{R_2} \\ &= 1.4 \times \frac{10}{14} = 1 \\ [b] &= 1 \end{aligned}$$

$$47. T_0 = 2\pi \sqrt{\frac{m}{6IB}} = 0.57 \text{ s}$$

For small oscillations:

$$\tau = I_{\text{mag}} B \sin \theta \approx I_{\text{mag}} B \theta$$

Magnetic moment:

$$I_{\text{mag}} = Ia^2$$

Moment of inertia of the square wire about OO_1 :

$$I_{\text{rot}} = \frac{ma^2}{6}$$

Thus,

$$T = 2\pi \sqrt{\frac{I_{\text{rot}}}{Ia^2 B}} = 2\pi \sqrt{\frac{m}{6IB}}$$

Putting $m = 0.01 \text{ kg}$, $I = 2 \text{ A}$, $B = 0.1 \text{ T}$:

$$\begin{aligned} T &= 2\pi \sqrt{\frac{0.01}{6(2)(0.1)}} \\ [T &\approx 0.57 \text{ s}] \end{aligned}$$

48. (5)

The magnetic moment of the N -turn wire is

$$\mu = NI\pi R^2$$

The magnetic torque is

$$\tau = \mu B = NI\pi R^2 B$$

For a solid sphere,

$$I_{\text{rot}} = \frac{2}{5}mR^2$$

Hence,

$$\alpha = \frac{\tau}{I_{\text{rot}}} = \frac{5NI\pi B}{2m}$$

Substituting $N = 10$, $I = 2 \text{ A}$, $B = 10 \text{ T}$, $m = 5 \text{ kg}$,

$$\alpha = \frac{5(10)(2)\pi(10)}{2(5)} = 100\pi \text{ rad/s}^2$$

Given

$$\alpha = 20n\pi$$

Therefore,

$$20n\pi = 100\pi$$



$$n = 5$$

49. (4)

Here $\vec{E} \parallel \vec{B}$, both along the x -axis.

The magnetic force only changes the direction of the y - z velocity, so

$$v_y^2 + v_z^2 = v_0^2$$

Electric field gives acceleration along x :

$$a_x = \frac{qE_0}{m}$$

When the speed becomes $2v_0$,

$$(2v_0)^2 = v_x^2 + v_0^2$$

$$v_x = \sqrt{3} v_0$$

Since $v_x = a_x t$,

$$\sqrt{3} v_0 = \frac{qE_0}{m} t$$

Therefore,

$$t = \frac{\sqrt{3} m v_0}{q E_0}$$

50. (4)

For a wire carrying current i in a uniform magnetic field,

$$\vec{F} = i(\vec{L} \times \vec{B})$$

The effective length $\vec{L}$ is the straight-line distance between the two ends of the wire.

Here,

$$L = 2b$$

and B is perpendicular to the plane of the wire.

Therefore,

$$F = i(2b)B = 2ibB$$

Hence acceleration,

$$a = \frac{F}{m} = \frac{2ibB}{m}$$

51. (1)

For $y > 0$, magnetic field is out of the page.

Initially,

$$\vec{v} = v\hat{j}, \vec{B} = B_1\hat{k}$$

For $q > 0$,

$$\vec{F} = q(\vec{v} \times \vec{B}) = q(v\hat{j} \times B_1\hat{k}) = qvB_1\hat{i}$$

So the particle bends towards $+x$, as shown. Hence,

$$q > 0$$

The radius of circular motion is

$$r = \frac{mv}{qB}$$

From the sketch, the radius in the $y > 0$ region is larger than that in the $y < 0$ region:

$$r_1 > r_2$$

Therefore,

$$\begin{aligned} B_1 &< B_2 \\ q > 0, |B_1| &< |B_2| \end{aligned}$$

52. $B = n \mu_0 I \tan(\pi n) / 2\pi R$, for $n \rightarrow \infty$ $B = \mu_0 I / 2R$

For one side of the regular n -sided polygon:

$$a = R \cos \frac{\pi}{n}$$

where a is the perpendicular distance from the centre.

The magnetic field due to one side at the centre is

$$B_1 = \frac{\mu_0 I}{2\pi a} \sin \frac{\pi}{n}$$

All n sides contribute equally, so

$$\begin{aligned} B &= nB_1 \\ B &= \frac{\mu_0 I n}{2\pi R} \tan \frac{\pi}{n} \end{aligned}$$

As $n \rightarrow \infty$

Using

$$\tan \frac{\pi}{n} \approx \frac{\pi}{n}$$

we get

$$B \rightarrow \frac{\mu_0 I}{2R}$$

So, the polygon approaches a circular loop.

53. $B = 4\mu_0 I / \pi d \sin \phi = 0.10 \text{ mT}$

Let the half-sides of the rectangle be a and b , and half-diagonal

$$R = \frac{d}{2} = 8 \text{ cm}$$

If the angle between diagonals is ϕ ,

$$a = R \cos \frac{\phi}{2}, b = R \sin \frac{\phi}{2}$$

The magnetic field at the centre due to all four sides is

$$B = \frac{\mu_0 I}{\pi R} \left(\cot \frac{\phi}{2} + \tan \frac{\phi}{2} \right)$$

Using

$$\cot x + \tan x = \frac{2}{\sin 2x}$$

we get

$$B = \frac{2\mu_0 I}{\pi R \sin \phi} = \frac{4\mu_0 I}{\pi d \sin \phi}$$

Substituting $I = 5 \text{ A}$, $d = 0.16 \text{ m}$, $\phi = 30^\circ$:



$$B = \frac{4(4\pi \times 10^{-7})(5)}{\pi(0.16)(0.5)}$$

$$\boxed{B = 1.0 \times 10^{-4} \text{ T}}$$

54. $B = (\pi - \phi + \tan \phi) \mu_0 I / 2\pi R = 28 \mu\text{T}$

Magnetic induction due to the arc segment at O ,

$$B_{\text{arc}} = \frac{\mu_0 i}{4\pi R} (2\pi - 2\phi)$$

and magnetic induction due to the line segment at O ,

$$B_{\text{line}} = \frac{\mu_0 i}{4\pi R \cos \phi} [2 \sin \phi]$$

So, total magnetic induction at O ,

$$B_o = B_{\text{arc}} + B_{\text{line}} = \frac{\mu_0 i}{2\pi R} [\pi - \phi + \tan \phi] = 28 \mu\text{T}$$

55. (a) $B = \frac{\mu_0 I}{4\pi} \left(\frac{2\pi - \phi}{a} + \frac{\phi}{b} \right)$;

(b) $B = \frac{\mu_0 I}{4\pi} \left(\frac{3\pi}{4a} + \frac{\sqrt{2}}{b} \right)$

(a)

For a circular arc,

$$B = \frac{\mu_0 I}{4\pi r} \theta$$

The two radial portions give zero field at O.

The two arcs produce opposite fields, hence

$$\boxed{B = \frac{\mu_0 I}{4\pi} \left(\frac{2\pi - \phi}{a} + \frac{\phi}{b} \right)}$$

(b)

The radial portions give zero field at O. Adding the contributions of the circular and straight portions,

$$\boxed{B = \frac{\mu_0 I}{4\pi} \left(\frac{3\pi}{4a} + \frac{\sqrt{2}}{b} \right)}$$

These are the required expressions.

56. $B = \mu_0 I / \pi^2 R$

Since the wire has a thin semicircular cross-section, the current I is uniformly distributed over the semicircle.

For an elemental current dI ,

$$dI = \frac{I}{\pi} d\theta$$

The magnetic field at O due to this element is

$$dB = \frac{\mu_0 dI}{2\pi R}$$

Only the horizontal components add; by symmetry, vertical components cancel.

$$B = 2 \int_0^\pi \frac{\mu_0 I}{2\pi R \pi} \sin \theta d\theta$$

$$B = \frac{\mu_0 I}{\pi^2 R}$$

Hence,

$$\boxed{B = \frac{\mu_0 I}{\pi^2 R}}$$

Direction is perpendicular to the plane of the cross-section.

57. (a) $B = (\mu_0 / 4\pi) (\pi I / R)$;

(b) $B = (\mu_0 / 4\pi) (1 + 3\pi/2) I / R$

(c) $B = (\mu_0 / 4\pi) (2 + \pi) I / R$

Using

$$B_{\text{arc}} = \frac{\mu_0 I \theta}{4\pi R}, B_{\text{straight}} = \frac{\mu_0 I}{4\pi R}$$

and applying the **right-hand rule**:

(a)

Straight parts pass through O, so their field is zero.

$$\theta = \pi$$

$$\boxed{B_a = \frac{\mu_0 \pi I}{4\pi R}}$$

(b)

$$\theta = \frac{3\pi}{2}$$

The straight part adds to the arc field:

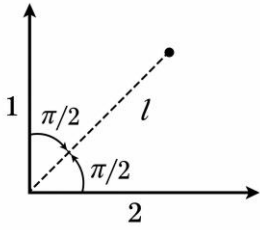
$$\boxed{B_b = \frac{\mu_0}{4\pi} \left(1 + \frac{3\pi}{2} \right) \frac{I}{R}}$$

(c)

Semicircular arc + two straight parts:

$$\boxed{B_c = \frac{\mu_0}{4\pi} (2 + \pi) \frac{I}{R}}$$

58. $B = (\mu_0/4\pi)I\sqrt{2}/l = 2.0 \mu \text{ T}$.



$$\vec{B}_0 = \vec{B}_1 + \vec{B}_2$$

$$B = \frac{\mu_0 i}{4\pi r} (\sin \theta_1 + \sin \theta_2),$$

using above equation

$$\text{or, } |\vec{B}_0| = \frac{\mu_0 i}{4\pi l} \sqrt{2} = 2.0 \mu \text{ T},$$

59. (a) $B = (\mu_0/4\pi) \sqrt{4 + \pi^2} I/R = 0.30 \mu \text{ T}$

(b) $B = (\mu_0/4\pi) \times \sqrt{2 + 2\pi + \pi^2} I/R = 0.34 \mu \text{ T}$

(c) $B = (\mu_0/4\pi) \sqrt{2} I/R = 0.11 \mu \text{ T}$

$$\begin{aligned} \vec{B}_0 &= \vec{B}_1 + \vec{B}_2 + \\ \vec{B}_3 &= \frac{\mu_0 i}{4\pi R} (-\vec{k}) + \frac{\mu_0 i}{4\pi R} \pi(-\vec{i}) + \frac{\mu_0 i}{4\pi R} (-\vec{k}) \\ &= -\frac{\mu_0 i}{4\pi R} [2\vec{k} + \pi\vec{i}] \end{aligned}$$

$$\text{So, } |\vec{B}_0| = \frac{\mu_0 i}{4\pi R} \sqrt{\pi^2 + 4} = 0.30 \mu \text{ T}$$

$$\begin{aligned} \text{(b) } \vec{B}_0 &= \vec{B}_1 + \vec{B}_2 + \vec{B}_3 \\ &= \frac{\mu_0 i}{4\pi R} (-\vec{k}) + \frac{\mu_0 i}{4\pi R} \pi(-\vec{i}) + \frac{\mu_0 i}{4\pi R} (-\vec{i}) \\ &= -\frac{\mu_0 i}{4\pi R} [\vec{k} + (\pi + 1)\vec{i}] \end{aligned}$$

So,

$$|\vec{B}_0| = \frac{\mu_0 i}{4\pi R} \sqrt{1 + (\pi + 1)^2} = 0.34 \mu \text{ T}$$

(c) Here using the law of parallel resistances

$$i_1 + i_2 = i \text{ and } \frac{i_1}{i_2} = \frac{1}{3},$$

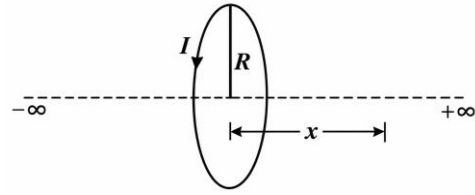
$$\text{So, } \frac{i_1 + i_2}{i_2} = \frac{4}{3}$$

$$\text{Hence } i_2 = \frac{3}{4}i, \text{ and } i_1 = \frac{1}{4}i$$

$$\text{Thus } \vec{B}_0 = \frac{\mu_0 i}{4\pi R} (-\vec{k}) + \frac{\mu_0 i}{4\pi R} (-\vec{j})$$

$$\begin{aligned} &+ \left[\frac{\mu_0}{4\pi} \left(\frac{3\pi}{2} \right) \frac{i_1}{R} (-\vec{i}) + \frac{\mu_0}{4\pi} \left(\frac{\pi/2}{R} \right) i_2 \vec{i} \right] \\ &= \frac{-\mu_0 i}{4\pi R} (\vec{j} + \vec{k}) + 0 \\ \text{Thus, } |\vec{B}_0| &= \frac{\mu_0 \sqrt{2} i}{4\pi R} = 0.11 \mu \text{ T} \end{aligned}$$

60. The given integral is equal to $\mu_0 I$.



The magnetic field at point P on the axis of a coil at a distance x from the center of the coil is:

$$B_x = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}$$

The axis of the loop taken along the x -axis:

$$\begin{aligned} \int_{-\infty}^{+\infty} \vec{B} \cdot d\vec{r} &= \int_{-\infty}^{+\infty} B_x dx = \frac{\mu_0 I R^2}{2} \int_{-\infty}^{+\infty} \frac{dx}{(R^2 + x^2)^{3/2}} \\ &= \frac{\mu_0 I R^2}{2} \int_{-\pi/2}^{+\pi/2} \frac{R \sec^2 \theta}{R^3 \sec^3 \theta} = \frac{\mu_0 I}{2} \int_{-\pi/2}^{+\pi/2} \cos \theta d\theta = \mu_0 I \end{aligned}$$

The integral on the LHS can be thought of as the line integral of B over a closed path (a straight line from $-\infty$ to $+\infty$ and a semicircle of infinite radius where the magnetic field vanishes). Then as per Ampere's law, this must be μ_0 times the current enclosed by the path which is I.

$$61. B = \begin{cases} 1/2\mu_0 [jr] & \text{for } r \leq R \\ 1/2\mu_0 [jr] R^2 / r^2 & \text{for } r \geq R \end{cases}$$

By circulation theorem inside the conductor

$$B_\phi 2\pi r = \mu_0 j_z \pi r^2$$

$$\text{or, } B_\phi = \mu_0 j_z r / 2$$

$$\text{i.e., } \vec{B} = \frac{1}{2} \mu_0 \vec{j} \times \vec{r}$$

Similarly outside the conductor,

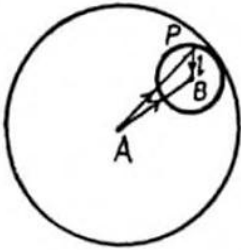
$$B_{\omega} 2\pi r = \mu_0 j_z \pi R^2$$

$$\text{or, } B_{\phi} = \frac{1}{2} \mu_0 j_z \frac{R^2}{r}$$

$$\text{So, } \vec{B} = \frac{1}{2} \mu_0 (\vec{j} \times \vec{r}) \frac{R^2}{r^2}$$

62. $B = 1/2 \mu_0 [jI]$, i.e., field inside the cavity is uniform

We can think of the given current which will be assumed uniform, as arising due to a negative current, flowing in the cavity, superimposed on the true current, everywhere including the cavity. Then from the previous problem, by superposition.



$$\vec{B} = \frac{1}{2} \mu_0 \vec{j} \times (A\vec{P} - B\vec{P}) = \frac{1}{2} \mu_0 \vec{j} \times \vec{l}$$

If vector I vanishes so that the cavity is concentric with the conductor, there is no magnetic field in the cavity.

63. $B = \begin{cases} \mu_0 jx & \text{inside the plate.} \\ \mu_0 jd & \text{out side the plate.} \end{cases}$

We assume that the current flows perpendicular to the plane of the paper, by circulation theorem,

$$2Bdl = \mu_0 (2xdl)j$$

$$\text{or, } B = \mu_0 xj, |x| \leq d$$

$$\text{Outside, } 2Bd = \mu_0 (2d * dl)j$$

$$\text{or, } B = \mu_0 dj, |x| \geq d$$

64. $p_m = 2\pi R^2 B / \mu_0 = 30 \text{ mA} \cdot \text{m}^2$

Magnetic moment of a current loop is given by $p_m = niS$ (where n is the number of turns and S , the cross sectional area.) In our problem, $n = 1$,

$$S = \pi R^2 \text{ and } B = \frac{\mu_0 i}{2R}$$

$$\text{So, } p_m = \frac{2BR}{\mu_0} \pi R^2 = \frac{2\pi BR^3}{\mu_0} = 30 \times 10^{-3} \text{ A} \cdot \text{m}^2$$

65. (a) $B = \frac{\mu_0 IN \ln(b/a)}{2(b-a)} = 7 \mu T$

$$(b) p_m = 1/3 \pi IN(a^2 + ab + b^2) = 15 \text{ mA} \cdot \text{m}^2$$

Yes, **your given answer is correct**. My numerical value was slightly off due to rounding/calculation.

Short proper solution

For a tightly wound spiral,

$$dB = \frac{\mu_0 I}{2r} dN$$

and

$$dN = \frac{N}{b-a} dr$$

Hence,

$$B = \frac{\mu_0 IN}{2(b-a)} \int_a^b \frac{dr}{r}$$

$$B = \frac{\mu_0 IN}{2(b-a)} \ln\left(\frac{b}{a}\right)$$

Putting $N = 100, I = 8 \times 10^{-3} \text{ A}, a = 0.05 \text{ m}, b = 0.10 \text{ m}$

$$B \approx 7 \mu T$$

(b) Magnetic moment

$$dp_m = I\pi r^2 dN$$

Therefore,

$$p_m = \frac{\pi IN}{b-a} \int_a^b r^2 dr$$

$$p_m = \frac{\pi IN}{3} (a^2 + ab + b^2)$$

$$p_m = 15 \text{ mA} \cdot \text{m}^2$$

66. (a) $B = 1/2 \mu_0 \sigma \omega R$

$$(b) p_m = 1/4 \pi \sigma \omega R^4$$

For a thin ring of radius r and width dr :

$$dq = \sigma(2\pi r dr)$$

Since the disc rotates with angular velocity ω , the ring current is



$$dl = \frac{dq}{2\pi/\omega} = \sigma\omega r dr$$

(a) Magnetic field at centre

Field due to the ring:

$$dB = \frac{\mu_0 dl}{2r} = \frac{\mu_0 \sigma \omega}{2} dr$$

Integrating from 0 to R ,

$$B = \frac{\mu_0 \sigma \omega R}{2}$$

(b) Magnetic moment

$$dm = dl \cdot \pi r^2$$

$$dm = \pi \sigma \omega r^3 dr$$

Therefore,

$$m = \int_0^R \pi \sigma \omega r^3 dr$$

$$m = \frac{\pi \sigma \omega R^4}{4}$$

So,

$$B = \frac{\mu_0 \sigma \omega R}{2}, m = \frac{\pi \sigma \omega R^4}{4}$$

$$67. p_m = 2\pi R^2 B / \mu_0 = 30 \text{ mA} \cdot \text{m}^2.$$

Magnetic moment of a current loop is given by

$p_m = niS$ (where n is the number of turns and S , the cross sectional area.) In our problem, $n = 1$,

$$S = \pi R^2 \text{ and } B = \frac{\mu_0 i}{2R}$$

$$\text{So, } p_m = \frac{2BR^2}{\mu_0} \pi R^2 = \frac{2\pi BR^3}{\mu_0} = 30 \times 10^{-3} \text{ A} \cdot \text{m}^2$$

$$68. B = (2\rho g S / I) \tan \theta = \text{mT, where } \rho \text{ is the density of copper.}$$

The Ampere forces on the sides OP and $O'P'$ are directed along the same line, in opposite directions and have equal values, hence the net force as well as the net torque of these forces about the axis OO' is zero. The Ampere-force on the segment PP' is effective and is deflecting in nature. In equilibrium (in the dotted position) the deflecting torque must be equal to the restoring torque, developed due to the weight of the shape.

Let, the length of each side be 1 and ρ be the density of the material then,

$$ilB (l \cos \theta)$$

$$= (Sl\rho)g \frac{1}{2} \sin \theta + (Sl\rho)g \frac{1}{2} \sin \theta + (Sl\rho)gl \sin \theta$$

$$\text{or, } il^2 B \cos \theta = 2 S \rho g l^2 \sin \theta$$

$$\text{Hence, } B = \frac{2 S \rho g}{i} \tan \theta$$

$$69. F_1 = \frac{\mu_0}{4\pi} \frac{2I_1 I_2}{b} \ln(1 + b/a)$$

The right-hand conductor carries current I_2 uniformly over width b .

Consider a small strip of width dx at distance x from I_1 .

$$dI_2 = \frac{I_2}{b} dx$$

Force per unit length between I_1 and dI_2 :

$$\frac{dF}{L} = \frac{\mu_0 I_1 dI_2}{2\pi x}$$

Therefore,

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi b} \int_a^{a+b} \frac{dx}{x}$$

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi b} \ln \left(\frac{a+b}{a} \right)$$

Since the currents are in the same direction, the force is attractive.

$$70. F_1 = \mu_0 I^2 / \pi^2 R$$

The magnetic field at distance R from the central straight conductor is

$$B = \frac{\mu_0 I}{2\pi R}$$

The current is uniformly distributed along the semicircular conductor:

$$dI = \frac{I}{\pi} d\theta$$

Force on an element per unit length:

$$\frac{dF}{l} = B dI$$

Hence,



$$\frac{F}{l} = \frac{\mu_0 I}{2\pi R} \frac{I}{\pi} \int_{\pi}^{2\pi} \sin \theta \, d\theta$$

$$\boxed{\frac{F}{l} = \frac{\mu_0 I^2}{\pi^2 R}}$$

The force is away from the central conductor (repulsive).

71. ($i\lambda B$)

For a current-carrying wire in magnetic field,

$$d\vec{F} = i(d\vec{l} \times \vec{B})$$

Here,

$$\vec{B} = B\hat{k}, \quad d\vec{l} = dx\hat{i} + dy\hat{j}$$

Thus,

$$d\vec{F} = iB(dy\hat{i} - dx\hat{j})$$

Integrating from $x = 0$ to $x = \lambda$:

$$\vec{F} = iB[(y_\lambda - y_0)\hat{i} - (\lambda - 0)\hat{j}]$$

Since

$$y_0 = y_\lambda = 0,$$

we get

$$\vec{F} = -iB\lambda\hat{j}$$

Hence the magnitude is

$$\boxed{F = iB\lambda}$$

72. $2iRB$, upward in the figure

For a current-carrying wire in a uniform magnetic field,

$$\vec{F} = i(\vec{L} \times \vec{B})$$

Only the portion of the wire inside the magnetic field experiences force.

The effective length of the immersed wire is the straight-line distance between its entry and exit points:

$$L = 2R$$

Since $\vec{B}$ is perpendicular to the plane of the wire,

$$F = i(2R)B$$

Hence,

$$\boxed{F = 2iBR}$$

The direction is towards the right by the right-hand rule.

73. ($4.9 \times 10^{-5} \text{ T}$)

$$\text{mass} = 10\text{mg} = 10^{-5}\text{Kg}$$

$$\text{length} = 1 \text{ m, } i = 2 \text{ A, } B = ?$$

$$\text{Now, } mg = i\ell B$$

$$\Rightarrow B = mg/i\ell = 4.9 \times 10^{-5} \text{ T}$$

74. $\frac{\mu mg}{il}$

The magnetic force on the wire is

$$F_m = BIl$$

The frictional force is

$$F_f = \mu N = \mu mg$$

For the wire to just start sliding,

$$BIl = \mu mg$$

Therefore, the minimum magnetic field is

$$\boxed{B_{\min} = \frac{\mu mg}{Il}}$$

where m is the mass of the wire.

75. (a) id/B towards the centre, (b) iab

(a) $F_{dl} = i \times dl \times B$ towards centre. (By cross product rule)

(b) Let the length of subtends an small angle of 2θ at the centre Here,

$$2T \sin \theta = i \times dl \times B$$

$$2T\theta = i \times a \times 2\theta \times B \quad [\text{As } \theta \rightarrow 0, \sin \theta \approx 0]$$

$$T = i \times a \times B$$

$$T = iaB$$

Force of compression on the wire = iaB

76. (a) id/B towards the centre, (b) iab

(a) $F_{dl} = i \times dl \times B$ towards centre. (By cross product rule)

(b) Let the length of subtends an small angle of 2θ at the centre Here,

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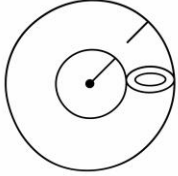
$$T = i \times a \times B$$

$$T = iaB$$



Force of compression on the wire = iaB

77. $\frac{ia^2B}{\pi r^2Y}$



$$Y = \text{stress} / \text{strain} = (F/\pi r^2)/(dl/L)$$

$$\Rightarrow \frac{dl}{L}Y = \frac{F}{\pi r^2} \Rightarrow dl = \frac{F}{\pi r^2} \times \frac{L}{Y}$$

$$= \frac{iaB}{\pi r^2} \times \frac{2\pi a}{Y} = \frac{2\pi a^2 iB}{\pi r^2 Y}$$

So, $dp = \frac{2\pi a^2 iB}{\pi r^2 Y}$ (for small cross sectional circle)

$$dr = \frac{2\pi a^2 iB}{\pi r^2 Y} \times \frac{1}{2\pi} = \frac{a^2 iB}{\pi r^2 Y}$$

78. $\sqrt{\frac{2qE_0z}{m}}$

Since the magnetic field does no work, only the electric field changes the kinetic energy.

Using the work-energy theorem:

$$qEz = \frac{1}{2}mv^2$$

The particle starts from rest, so

$$v^2 = \frac{2qEz}{m}$$

Hence the speed as a function of z is

$$v(z) = \sqrt{\frac{2qE_0z}{m}}$$

(for $qE_0z > 0$).

79. (*)

Electric field between the plates:

$$E = \frac{V}{d}$$

Electric force on electron:

$$F = eE$$

Hence,

$$a = \frac{eE}{m_e}$$

Starting from rest and moving through distance d :

$$v^2 = 2ad$$

$$v^2 = 2 \frac{eV}{dm_e} d$$

$$v = \sqrt{\frac{2eV}{m_e}}$$

In the magnetic field, the electron moves in a circular path of radius

$$r = \frac{m_e v}{eB}$$

Therefore,

$$r = \frac{m_e}{eB} \sqrt{\frac{2eV}{m_e}} = \frac{\sqrt{2m_e V/e}}{B}$$

The electron will fail to strike the upper plate if

$$d > r$$

Thus,

$$d > \sqrt{\frac{2m_e V}{eB^2}}$$

80. (0.5 T)

Given:

$$N = 100, l = 5 \text{ cm} = 0.05 \text{ m}, b = 4 \text{ cm} = 0.04 \text{ m}$$

$$I = 2 \text{ A}, \tau = 0.2 \text{ Nm}$$

Since the coil's plane is parallel to B , its normal is perpendicular to B . Thus,

$$\tau = NIAB$$

where

$$A = lb = (0.05)(0.04) = 2 \times 10^{-3} \text{ m}^2$$

Therefore,

$$B = \frac{\tau}{NIA}$$

$$B = \frac{0.2}{100 \times 2 \times 2 \times 10^{-3}}$$

$$B = 0.5 \text{ T}$$



81. (a) $6.3 \times 10^{-2} \text{ N m}$, (b) 60°

$$n = 50, r = 0.02 \text{ m}$$

$$A = \pi \times (0.02)^2, B = 0.02 \text{ T}$$

$$i = 5 \text{ A}, \mu = niA = 50 \times 5 \times \pi \times 4 \times 10^{-4}$$

T is max. when $\theta = 90^\circ$

$$\begin{aligned} T &= \mu \times B = \mu B \sin 90^\circ = \mu B \\ &= 50 \times 5 \times 3.14 \times 4 \times 10^{-4} \times 2 \times 10^{-1} \\ &= 6.28 \times 10^{-2} \text{ N - M} \end{aligned}$$

Given $T = (1/2)T_{\text{max}}$

$$\Rightarrow \sin \theta = (1/2)$$

or, $\theta = 30^\circ$ = Angle between area vector & magnetic field.

$\Rightarrow$ Angle between magnetic field and the plane of the coil $= 90^\circ - 30^\circ = 60^\circ$



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>